# BEVY_ZEROVERSE: AUDITABLE PROCEDURAL INDOOR DATA FOR MULTI-VIEW RECONSTRUCTION

A PREPRINT

**Mitchell Mosure**
mitchell@mosure.me

October 2026

**ABSTRACT**

We describe an open-source synthetic data engine combining primitive-object scenes with furnished procedural interiors, calibrated multi-view cameras, optional human motion, and aligned geometric annotations. Indoor generation samples continuous architectural, furnishing, appearance, lighting and camera parameters under geometric constraints. A continuous camera-baseline control coordinates spacing, shared-surface overlap and path variation. The architectural evaluation audits 512 consecutive rooms and 256 aligned views across 32 rooms. A matched camera study covers 64 rooms, 1–12 cameras and 3 baseline values, with native shared-surface measurements from 2 prespecified room families. Exact configurations, capture-engine identities and reproducible artifacts accompany the measured cohorts. Photographic realism, unlimited-process stability and downstream pretraining gains are not established by these diagnostics.




Figure 1: Current generated architecture: mezzanine stairs, a sunken lounge and a raised platform with an arched portal. Native RGB at $t = 0$; seeds 7, 8 and 6, cameras 3, 3 and 1. Plans, calibrated views, aligned annotations and the unfiltered cohort follow in the main evaluation.

## 1 Purpose and scope

Sparse-view reconstruction benefits from observations of shared surfaces: cameras that see unrelated objects offer little direct correspondence supervision. LRM-Zero demonstrates the potential of procedurally synthesized object data [1]; MVSplat uses cross-view matching to estimate geometry [2]. Our contribution is an inspectable data-generation system and its bounded evaluation, rather than a trained reconstruction model or a reconstruction accuracy result.

The engine supports Object, CornellCube, Room, SemanticRoom and Human scene modes alongside furnished interiors in `ProceduralIndoor`. The runtime uses Bevy 0.20, optional Burn 0.22 inference, and published AnnyBody/ARDY integrations. Office, conference, lounge, classroom, library and related activity priors influence a compositional program; they do not select complete meshes or a finite collection of room templates. Most architecture, furniture,

plants and clutter require neither mesh repositories nor photographic texture collections. AnnyBody human meshes and optional motion/embedding models are explicit pretrained dependencies.

## 2 Procedural scene program

A scene is generated as $S = G(s, \theta)$, where $s$ seeds independent random streams and $\theta$ contains user-controlled densities, activity, quality, camera and motion policies. The serialized manifest records the sampled program, object transforms and seeds, people, photometry and camera paths. A separate generator version and capture-engine identity invalidate incompatible resumptions. Determinism means reproducing the same program and settings with compatible assets and engine versions; bit-identical rasterization across GPU vendors is not promised.

### 2.1 Architecture, objects and appearance

Room area and aspect, functional zones, aisle widths and activity mixtures are sampled continuously. The exterior footprint supports taper, chamfered corners and a concave cut-in. A shared architectural program controls polygon triangulation, wall apertures, sloping ceilings, arched portals, pillars, floor levels and furnished mezzanines with stairs and guards. Furniture support, camera clearance, fixture anchors and voxel export use the same surfaces. Windows vary aperture dimensions, sill/head heights, bay widths, frames, inset depth and shades across multiple exterior edges. An internal glazed connection retains a furnished neighboring room. This is a bounded architectural grammar: courtyard cut-ins remain on the rear facade, neighboring rooms and mezzanine decks remain rectangular, and arbitrary multi-storey connectivity, structural analysis and building-code compliance are outside scope.

Furniture is assembled from material-specific parts. Tables vary outline, taper, thickness and leg/pedestal construction; chairs vary seat shell, curvature, back proportions, recline, arms, base and headrest. Sectional seating, shelving, potted plants, outlets, switches, ceiling lights, whiteboards, displays and tabletop objects extend the semantic inventory. Placement uses support relationships, clearance checks, zone-dependent furnishing fields and bounded rejection. Chairs sample partial and deep insertion beneath tables, including occupied seats; the rendered leg, pedestal, brace and panel program also supplies collision geometry. Working poses mix free reaches with one- or two-palm tabletop support using fixed-length arm IK and skin-aware contact fitting. This is static surface support, not simulated grasping or animated contact dynamics. Surface clutter varies shared arrangement, handedness, spacing and local jitter. Object yaw and arrangement are sampled rather than restricted to orthogonal grids. Collision proxies approximate detailed meshes; geometric validity is checked separately from physical plausibility.

Surfaces use 36 role-specific base material slots with procedural albedo, normal and roughness maps, metric UV scales and mip chains. Recipes vary spatial frequency, directional grain, textile structure, mineral veining, relief, wear, color and roughness. Upholstery finish slots share three independently seeded structure programs per substrate; clothing reuses these maps with independent actor pigmentation and atlas scaling. Woven textiles sample over/under grammars, yarn diameter, slub, crimp, dyed warp/weft and fuzz. Leather varies grain spacing, stretch, pore relief, nap, pigmentation and polish. Coupled albedo, metric normal, cavity and roughness fields are baked into mipmapped textures. Textile directional GGX scattering approximates aggregate fibre reflection; it is not a measured cloth multiple-scattering BSDF. Mineral programs blend cement binder, exposed aggregate, pinholes and warped multiscale vein deposits with independent polish. Wall finishes vary roller stipple, sprayed and flattened plaster relief, trowel marks and gloss; glazed ceramics add restrained crackle and a protective lobe. Timber separates grain cuts from stain absorption, bleaching, pore fill and varnish; face and edge grain share their finish. Foliage atlases keep veins aligned to the blade fold and vary chlorophyll patches, pale variegation, wax and thin-surface transmission. Timber and foliage finish slots share two texture structures per role. Timber and stone floor atlases use 512-square maps to resolve separately cut boards and stone tiles; other substrates use 256-square maps. Screens, boards and printed pages have independently seeded content. Only referenced finish variants are built during runtime preparation. Glass recipes vary roughness, refractive index, tint, attenuation and thickness. Direct lights vary placement, intensity, color temperature, circuit activity and direction; exposure, daylight and sky radiance are independently recorded. Low-light samples are retained in evaluation.

Timber samples offset and tapered growth-ring cuts, uneven earlywood/latewood, elongated pores and sparse branch knots; floor planks have independent cuts. Unresolved normal variation contributes to filtered GGX roughness. Chrome retains a conductive base layer with continuously varied polished/brushed roughness and anisotropy; micron scratches are represented as scattering rather than quantized visible bumps. A static scene-derived HDR environment traces the rendered triangle assemblies and visibility-tested direct lights using the shared transport BVH. Linear radiance is stored in float16 cubemaps, with GGX specular and cosine diffuse convolution; source footprints suppress sparse-emitter filtering noise. Secondary bounce energy is approximate. One reflection origin does not provide per-object parallax, live moving-human reflections or a full specular path-tracing reference.

People retain AnnyBody geometry with continuous phenotype/pose parameters, fitted garments, surface hair and facial details. Static people remain the default. Opt-in ARDY motion combines seeded procedural text sequences with object-aware waypoint planning and scene-time playback. Requesting static scenes does not initialize motion inference. Motion models use the upstream loaders and caches; cached prompt text does not replace the motion seed. Clothing, hair, joint limits and collision handling are approximations, particularly during complex motion.

### 2.2 Multi-view sampling and training defaults

Camera centers and complete trajectories are constrained to the primary furnished room by default. The sampler varies focal length, position, target, roll and path shape, checks obstacle clearance and prefers visible content. Static capture uses one timestep; optional trajectories support repeated calibrated observations. A reference camera anchors a connected set of views. For each other camera and five trajectory times, we require

$$\min\{o_{0\to i}(t), o_{i\to 0}(t)\} \geq \tau, \qquad b_{\min} \leq \|c_i(t) - c_0(t)\| \leq b_{\max},$$

where $o$ estimates first-surface overlap using visibility rays and object/architecture proxies. Overlap constrains reference edges. An additional minimum separation applies to every camera pair at 33 synchronized path samples. Group spread is the horizontal camera-center covariance's minor/major standard-deviation ratio. Moving pairs must differ in normalized RMS displacement after removing their starting offsets. Full-path swept collision checks remain; sampled spread and overlap are not continuous-time or rendered-pixel guarantees.

The camera-baseline control $b \in [0, 1]$ expands into explicit advanced settings: minimum pair separation $0.04 + 0.46b$ metres, minimum reference distance $0.06 + 2.30b$ metres, maximum reference distance $0.35 + 9.65b$ metres, overlap target $0.65 - 0.50b^{0.7}$, group spread $0.20 + 0.10b$, independent path variation $b$, and visible-region target variation $b$. The default is $b = 0.5$. Wide settings permit distant viewpoints and lower overlap; narrow settings favor close views of shared content. Collision and visibility rejection affect the realized distribution. Arbitrary room, camera-count and constraint combinations need not be feasible.

Peers can aim at different first-surface proxy hits visible to the reference camera. Target interpolation and proposal frequency increase with baseline; frequency also increases with camera count. Feasible candidates are ranked for nearest-peer distance, optical-axis separation and additional quantized primary-room surface cells. Preferred distance scales with room diagonal divided by the square root of camera count; it is not a hard packing constraint. Selection considers at most 2, 3 or 4 feasible candidates for groups of 2–4, 5–8 or 9+ views, within a 12-proposal window after the first feasible candidate and the existing total search budget. Zero target variation retains shared-target, first-fit rigs. Rare completion for 2–11 views can use one bounded overcomplete search, retaining the reference and rechecking every subset against the same geometric constraints. All selected views still satisfy the original collision, content, distance and overlap gates. Proxy surface-cell novelty does not measure exact mesh area or reconstruction gains. Related temporal routes and occasional shared-target recovery remain supported; varied gaze does not imply independent route topology.

The inspector exposes this slider above the advanced controls. Any advanced edit switches to explicit custom values. Changes take effect on regeneration, so dragging a control does not rebuild the room. JSON input accepts `{"baseline":0.75}` within the `indoor_camera` configuration. Serialization records the expanded values, avoiding hidden preset precedence. Camera travel length and long-route frequency are separate from spacing between cameras. Joint collision and overlap rejection can favor shorter accepted routes at wide spacing; the minimum travel bound remains explicit. The finite search budget reports impossible settings without silently weakening them.

## 3 Capture and annotation contract

Capture begins only after scene preparation, assets, shader pipelines, lighting and requested human trajectories are ready. A bounded asynchronous readback protocol associates outputs with the requested scene, camera, time and mode; it does not infer readiness from an arbitrary number of frames. Sequential native indoor streams automatically prepare future rooms on a shared bounded CPU pool and can upload one future room's immutable assets during current readback. Future entities, lighting dispatches and motion inference remain gated by scene promotion. This scheduling retains full native quality rather than reducing shadows or geometry. Geometric channels share an RGBA32Float ground-truth raster attachment. RGB uses Bevy's PBR pipeline, shadow maps, tone mapping and optional indirect-lighting approximation. Float32 storage does not imply that all RGB or lighting computations have float32 intermediate precision: HDR color and irradiance volumes include float16 stages.

Headless generation supports chunked safetensors or per-sample files, compression, resumable identity checks, metrics and bounded worker replacement. RGB and numeric channels have separate encoding contracts. Native generation performs numeric temporal export and O-Voxel export. The WebGPU viewer supports indoor construction, inspection

| Indoor training configuration | Default and rationale |
|---|---|
| Capture cameras / timesteps | Four / one; synchronized views of static geometry |
| Shared-view policy | $b = 0.5$: $\tau \approx 0.342$, pair separation $\geq 0.27$ m, reference distance 1.21–5.175 m; explicit `multiview:null` enables independent views |
| Group diversity | Horizontal spread ratio $\geq 0.25$ for 3+ views; moving-pair normalized RMS difference $\geq 0.075$ |
| Room/activity/density | Primary room, Mixed activity, furnishing density 0.65, human density 0.25 |
| Camera path policy | 0.03–8 m allowed path length, 0.45 long-path proposal fraction; collision rejection still applies |
| Human motion | Disabled; static people remain, no ARDY initialization |
| Native lighting | Auto quality, shadows and approximate baked diffuse GI, 256 rays/probe |
| Dataset CLI workers / chunks | One process, 16 rooms/chunk; explicit overrides retained |
| Finite worker lifetime | 256 scenes/child for finite indoor jobs with process workers |
| Default export | RGB and calibration; geometric channels are opt-in |

Table 1: Defaults conditional on the procedural-indoor dataset CLI scene. Other scene modes retain their defaults. These are an engineering starting point, not an optimum established by downstream training. Library callers explicitly choose camera and worker counts.

| Output | Meaning and qualification |
|---|---|
| Color | Display RGB; lossless float storage or explicitly chosen codecs. Separate linear-HDR export supports renderer comparisons. |
| Depth | Colorized, normalized or metric linear depth; axial/ray conventions; calibration records the selection. |
| Normal / position | View-space normals encoded as $(n + 1)/2$; world positions affine-normalized by the exported AABB, without clamping. First surface, float32 geometric path. |
| Semantic | Shared class vocabulary, aligned with depth/normal/position, including ceiling light fixtures. |
| Optical flow | Forward displacement $(\Delta u, \Delta v)$ in pixels, with correspondence-valid and target-visible masks. Terminal time has no forward correspondence. |
| Motion vectors | The same exported displacement divided by image width/height. Viewer velocity colors are previews, not these numeric annotations. |
| Co-visibility | Optional same-time, per-pixel membership in other capture cameras. A uint16 mask assigns one bit per camera (up to 16), with a separate source-valid mask and a consistent additive RGB legend; the source camera and editor are excluded. |
| Calibration / instances | Camera intrinsics, poses and normalized trajectory time, scene manifest, semantic instance oriented boxes, person joints, and primary-room annotation AABB. |
| O-Voxel | Primary-room architecture/objects only; excludes neighboring/outside geometry. Requires one timestep and disabled human motion; incompatible requests are rejected. |

Table 2: Eight currently supported render modes plus structured annotations. The architectural cohort exports six modes, including production co-visibility; optical flow and motion vectors are available for temporal dataset export but are not measured in this cohort. Glass is a first geometric surface by default. The opt-in through-glass policy excludes transmissive hits consistently from geometric, semantic and co-visibility annotations along unrefracted rays; RGB transmission is unchanged. Flow transports surface correspondence, including supported human deformation, rather than estimating photometric image motion.

and optional motion inference, using the same manifest and geometry modules; Portable quality reduces unsupported or costly rendering features. Browser velocity visualization is not a qualification of native dataset flow export. The current Wasm path omits native baked diffuse GI and uses its platform lighting fallback.

### 3.1 Co-visibility representation and surface agreement

Co-visibility shares the float32 surface raster used by the other geometric annotations. For source camera $s$, pixel $p$ and time $t$, the exported unsigned 16-bit integer is

$$M_s(p,t) = \sum_{\substack{j=0 \\ j \neq s}}^{N-1} 2^j V_{s \to j}(p,t), \qquad 1 \leq N \leq 16,$$

where $V_{s\to j} \in \{0,1\}$ records visibility of the source surface in peer camera $j$. The source camera's own bit is always zero; the editor has no membership bit. The ordered camera legend remains identical across views and timesteps in a sample. A peer's binary supervision is recovered by testing $(M_s \,\&\, 2^j) \neq 0$; the number of observing peers is $\mathrm{popcount}(M_s)$. Thus the integer preserves camera identities, unlike a shared-pixel count alone.

The GPU pass projects a valid source surface into each peer, checks near/far clipping and image bounds, and reads the nearest target pixel. For source point $x$, target point $y$, corresponding world-space unit normals $n_s, n_j$ and target axial depth $d$, the surface-agreement test is

$$\max(|(y-x)\cdot n_s|, |(y-x)\cdot n_j|) \leq \max(0.001\,\mathrm{m}, 10^{-4}d).$$

This local-plane test avoids comparing depths along different pixel-center rays on sloped surfaces. It remains resolution- and tolerance-dependent: silhouettes, thin geometry, curved surfaces and near-contact surfaces can be ambiguous. Pixel-level membership need not be symmetric after raster reprojection. The chosen solid or through-glass annotation policy applies to the source and every peer. Reflected and refracted RGB content does not imply co-visible geometry.

The additive RGB8 preview sums the contributions of set camera bits without tone mapping. For four cameras, the shared legend is $C_0 = (170, 0, 0)$, $C_1 = (0, 255, 0)$, $C_2 = (0, 0, 255)$ and $C_3 = (85, 0, 0)$. For example, $M = 10 = \texttt{0b1010}$ means peers 1 and 3, with color $(85, 255, 0)$. These colors encode camera membership, not semantic classes or overlap magnitude. The palette adapts deterministically to camera count and remains uniquely decodable for all supported combinations in the original lossless PNG.

A separate uint8 source-valid mask $A_s$ distinguishes an unshared surface ($A_s = 1, M_s = 0$) from background ($A_s = 0, M_s = 0$); both appear black in the additive preview. Folder exports retain membership and validity as $[H, W, 1]$ arrays in NPZ. Chunked exports use $[B, T, N, H, W, 1]$. Training should consume these numeric arrays and the camera metadata, rather than rasterizing the PDF or decoding a resized or lossy preview. Shared percentages divide by valid source-pixel observations, including unshared surfaces, not unique 3D points.

## 4 Procedural pipeline and measured cost

Figure 2 follows the production path from seeded programs to completed views. Immutable architecture, furnishing and person assemblies are constructed once. The same triangles supply transport and rendering. A shared pool with at most four dedicated preparation threads executes independent builders; scoped callers can also execute queued work. Large BVH subtrees use a separate shared pool of up to four threads. Ordered results and reserved mesh handles preserve seed identity. Material preparation overlaps geometry transport, group planning and mesh/tangent realization. Only referenced material variants are prepared. Supported material maps and their mip chains are baked once on the GPU; remaining programs use the CPU oracle. Native room lookahead overlaps preparation, bounded speculative uploads and static lighting with current capture, without promoting future scene entities early. The browser uses cooperative yields in place of native worker parallelism; yields keep the event loop responsive but do not provide CPU parallelism.

### 4.1 Protocol and accounting

Measurements use an Intel Core i9-12900K (24 logical CPUs) and NVIDIA RTX PRO 6000 Blackwell Workstation Edition using Vulkan. Rust 1.100.0-nightly (2026-08-31 toolchain), native x86-64 Linux, optimized dev profile (opt-level 3; debug information disabled), multi_threaded with default features disabled. Existing Gekko training and capture processes remained active; no other local qualification or compiler ran during the measured timing protocol. Device telemetry is board-wide and includes those other processes.. These are shared-machine measurements, not isolated hardware limits or a training-throughput benchmark. The receipt binds source SHA256 `8b9386343019c7a1`, package versions, exact configuration, input hashes and every measured seed. No geometry, texture resolution, shadow quality or GI-ray budget is reduced for this study. Static AnnyBody people are sampled at density 0.25; ARDY inference is disabled.

The CPU-only study samples 64 consecutive rooms, seeds 400–463, with 4 separate warm-up rooms. It invokes the same preparation entry point as capture, including camera sampling, layout validation, geometry, material recipes, transport, reflections and lighting requests. It prepares GPU requests without executing them and cannot measure rendered throughput. Table 3 partitions one future's monotonic wall lifetime into disjoint intervals. Values include scheduling/waiting within a stage, not CPU core time. Relative shares divide summed stage durations by summed preparation lifetimes; percentiles use nearest rank. Concurrent branch times and different rooms must not be summed into this partition.

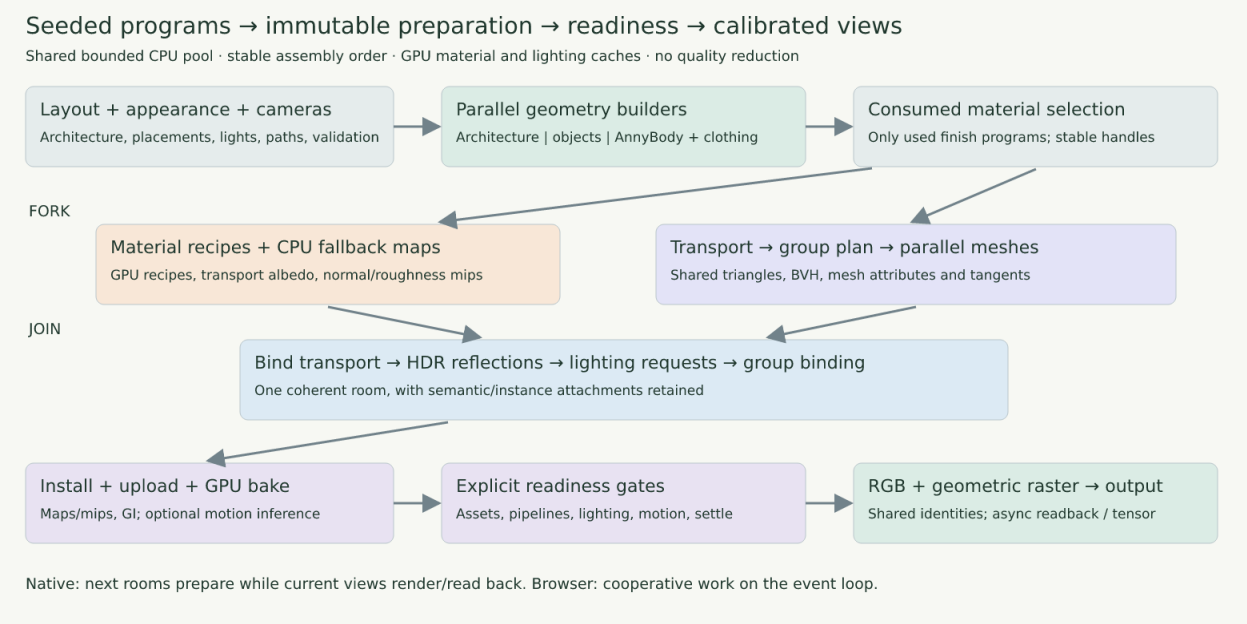



Figure 2: Production dependencies. The fork/join overlaps CPU material work with transport and mesh realization; GPU maps and lighting finish behind explicit readiness gates. Optional human inference is a separate gate. Consecutive-room lookahead spans these stages; the layout does not imply serial execution of whole rooms.

| Preparation phase | Mean (ms) | Median (ms) | p95 (ms) | Share (%) |
|---|---|---|---|---|
| Layout and appearance | 2.28 | 1.70 | 6.03 | 0.26 |
| Camera paths and overlap | 88.56 | 43.53 | 303.98 | 10.04 |
| Layout validation | 10.09 | 8.71 | 21.05 | 1.14 |
| Motion planning | 0.00 | 0.00 | 0.00 | 0.00 |
| Geometry assemblies | 46.06 | 43.66 | 77.68 | 5.22 |
| Consumed material selection | 0.14 | 0.12 | 0.23 | 0.02 |
| Materials / transport / meshes | 543.58 | 499.47 | 871.16 | 61.60 |
| Transport material binding | 7.14 | 6.99 | 11.29 | 0.81 |
| HDR reflection environment | 148.67 | 131.76 | 266.23 | 16.85 |
| Lighting bake preparation | 32.07 | 30.00 | 62.58 | 3.63 |
| Serial mesh tail (Wasm) | 0.00 | 0.00 | 0.00 | 0.00 |
| Render group binding | 0.18 | 0.16 | 0.27 | 0.02 |
| Scheduling and bookkeeping | 3.66 | 3.67 | 6.93 | 0.41 |
| Total | 882.44 | 804.79 | 1485.02 | 100.00 |

Table 3: CPU preparation with production GPU material recipes, three square-aspect cameras and native Auto quality. The material/transport/mesh row counts the union once. Asset installation, actual GPU bakes, rendering and readback are downstream.

### 4.2 Completed capture and bounded concurrency

Each native capture run regenerates rooms and delivers three 512×512 views at one timestep in RGB, depth, position, normal, semantic and co-visibility modes. Full native shadows, SSAO, GPU materials and 256-ray/probe GI remain enabled. The default three-room lookahead is used. The streaming denominator includes validation, diagnostic JSONL writes and asset/sample destruction: preparation can advance during these operations. No image compression, dataset encoding or Burn tensor handoff is measured. Capture latency covers request to complete sample; it differs from streaming wall cost. GPU timestamp instrumentation is disabled in throughput runs.

| Workload | Rooms | Wall rooms/s | Median (ms) | p95 (ms) |
|---|---|---|---|---|
| CPU preparation, 1 in flight | 64 | 1.129 | 804.8 | 1485.0 |
| CPU preparation, 4 in flight | 64 | 2.324 | 1570.9 | 2519.6 |
| Completed capture, run 1 | 64 | 0.995 | 964.2 | 1734.3 |
| Completed capture, run 2 | 64 | 1.081 | 839.9 | 1980.7 |

Table 4: Different denominators: CPU-only preparation emits immutable asset plans; completed capture emits all requested planes. Repeated capture runs expose shared-machine variability. Neither rate includes downstream training.

Capture run 1 excludes 16 warm-up rooms; setup took 1.71 s and the first room took 5.71 s. The measured stream delivered 2.99 views/s; capture spans occupied 99.27% of stream wall time. These spans overlap future-room work. Main-world installation records deferred ECS commands and asset insertion, not their later execution or upload.

Capture run 2 excludes 16 warm-up rooms; setup took 2.02 s and the first room took 5.16 s. The measured stream delivered 3.24 views/s; capture spans occupied 99.15% of stream wall time. These spans overlap future-room work. Main-world installation records deferred ECS commands and asset insertion, not their later execution or upload.

### 4.3 Hot spots and next optimization targets

The largest measured preparation windows are Materials / transport / meshes, HDR reflection environment and Camera paths and overlap (Figure 3). They are observations of this workload, not universal cost ratios. Within the parallel asset window, mean material preparation is 306.9 ms, geometry transport construction is 349.2 ms, and mesh/tangent realization is 181.8 ms. Transport precedes meshes; that branch overlaps material work. Geometry builder service times are separately retained by architecture, object and person categories; their sums are not a CPU-utilization measurement.

Priority candidates are reducing triangle-staging and BVH-construction memory traffic, improving scheduling of the existing parallel transport subtrees, and GPU or batched reflection-environment synthesis. Camera visibility searches need independent rejection/attempt profiling before changing their budget: reducing the search budget can bias the camera distribution. Geometry assembly is separately measured, so finer-grained person/architecture scheduling should be pursued only when it is on the critical path. Any move of map or reflection work to the GPU must retain deterministic inputs, normal/roughness mip correctness, annotation alignment and measured completed capture throughput while training competes for the device. Pool expansion can increase contention; more workers are not automatically faster. Broad caching keyed only by material category would reduce procedural diversity and is not an acceptable optimization.

Render-preparation, GI/material readiness, settling and readback event-loop buckets are retained in the receipt. A bucket names the blocker observed before an update; it is not an exclusive CPU or GPU execution timer. Asynchronous GPU diagnostics can arrive late and contain nested intervals, so they are not summed into per-room GPU time. A GPU timeline profiler is still needed to attribute submission gaps, uploads and individual kernels. This study does not measure isolated GPU utilization, device-to-device tensor throughput, motion inference or browser latency. Wasm compilation checks preserve the path; browser timing requires a separate hardware-WebGPU run.

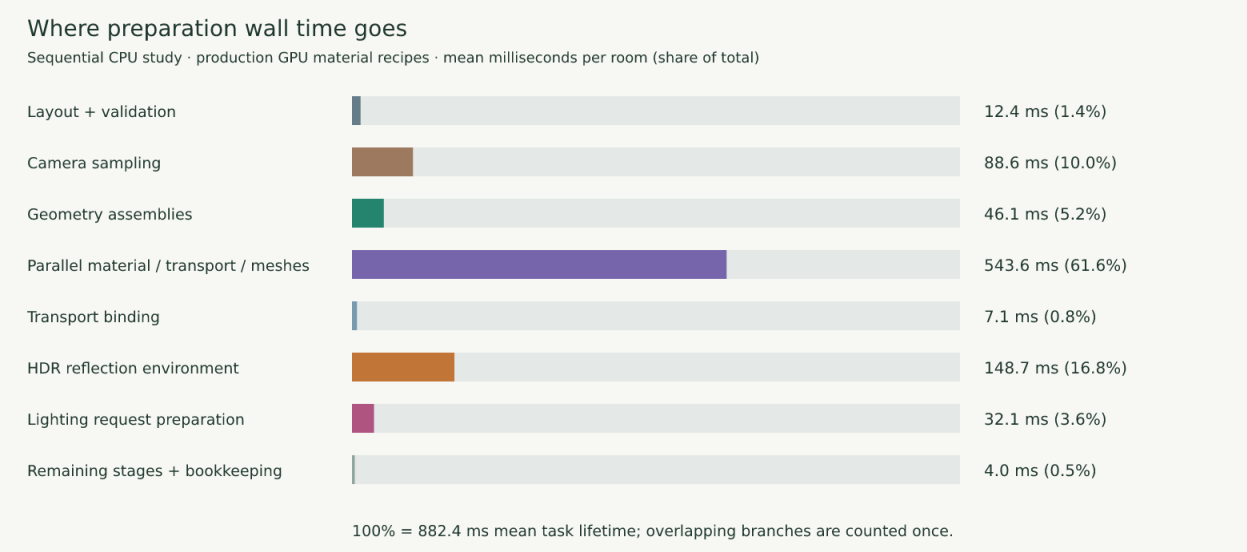



Figure 3: Mean per-room CPU preparation wall time and its relative partition for the sequential CPU study. Colors separate the major phases; the parallel window includes scheduling and both branches, exactly once. This is not a utilization chart.

# 5 Architectural evaluation

## 5.1 Protocol and architectural audit

This cohort was captured with bevy_zeroverse 0.40.0, generator 40, on 512 consecutive seeds (0–511), with 4 cameras per room, furnishing density 0.65 and static human density 0.25. Every audited room passed the recorded layout, support and camera-clearance checks. The first 32 rooms were rendered without appearance filtering at 640×400, with 2 trajectory endpoints: 256 camera/time identities in six aligned modes. The audit contains 512 unique structural signatures after centimetre quantization; this bounded diagnostic does not estimate the total procedural space or learning utility.

| Built feature | Rooms | Fraction |
|---|---|---|
| Arched portals | 194 / 512 | 37.9% |
| Chamfered corners | 178 / 512 | 34.8% |
| Exterior cut-in | 403 / 512 | 78.7% |
| Interior pillars | 422 / 512 | 82.4% |
| Mezzanine | 42 / 512 | 8.2% |
| Oblique walls | 497 / 512 | 97.1% |
| Raised floor | 94 / 512 | 18.4% |
| Sloping ceiling | 354 / 512 | 69.1% |
| Sunken floor | 99 / 512 | 19.3% |

Table 5: Architectural features in the complete consecutive population. Features co-occur; fractions do not sum to 100%.

## 5.2 Geometry, calibration and displayed captures

Footprint areas span 34.27–257.10 $m^2$, with roof pitch 0.00–11.32°. Figure 4 and Figure 5 pair native RGB with the actual serialized envelope plans and sections. Camera arrows indicate direction, rather than frusta. Solid furnishings are footprint proxies; the sections show roof and floor levels and omit furnishings, pillars and partitions. Structural feature examples are selected independently of rendered appearance; all rooms remain available in the contact sheet and explorer. Metric constraints are not building-code certification.

The page presents all 32 rooms, all four cameras and both endpoints in one explorer, with 1536 channel images. RGB, depth, normals, semantics, positions, co-visibility, camera calibration and plans share the same room/time identity. RGB preserves renderer tone mapping and sRGB encoding without per-image exposure correction or cropping. All previews use lossless WebP. Depth is an 8-bit axial-depth preview divided by 15 m and clamped; normals store $(n + 1)/2$, and positions are affine-normalized by the annotation AABB and clamped. Semantic colors are preserved exactly. These previews are not float32 training labels. Geometric annotations use the first surface, including opaque glass, rather than reflected or refracted optical content.

The native capture protocol uses Auto quality, shadows, SSAO and approximate diffuse GI with 1024 primary rays per probe. Renderer identity and capabilities are recorded in every calibration download. All captured views passed the recorded alignment checks: maximum per-view depth/position p99 error 2.861 $\mu$m, reprojection p99 0.00042 pixels and normal-length error 2.980e-7. Every view contains at least 7 semantic classes. 17/17 rooms with humans in the camera's primary functional zone show person pixels; 4/23 rooms with humans anywhere in the main envelope never show person pixels. These are class-level observations, rather than instance-level visibility guarantees.

Scene construction through capture/export takes 1.69–5.85 seconds per room (median 3.71), excluding application startup. This is a bounded capture run, not a sustained-throughput, unlimited-process memory-stability or photographic-realism qualification.

## 5.3 Co-visibility in every captured room

Production co-visibility accompanies every RGB and geometry view in this cohort. Each valid source pixel stores an exact uint16 membership mask of other capture cameras that see its first surface, with the source bit excluded. One deterministic additive legend is shared across all views. The unified explorer supports all memberships and individual-peer overlays; the selected peer's own tile stays RGB.

Across every room and endpoint, 83.2% of valid source-pixel observations are visible to at least one peer. This pixel-weighted statistic pools 65536000 valid source-pixel observations; it does not count unique 3D points. Exact

downloads include 16-bit grayscale membership PNGs, separate 8-bit 0/1 validity PNGs, calibrated transforms and the ordered camera legend. Zero membership with valid=1 is an unshared surface; zero membership with valid=0 is background. Co-visibility and other annotations were captured together, rather than joined from separate runs.

### 5.4 Matched camera and appearance qualification

A separate prespecified factor sweep covers 2 geometry families, eleven settings, two cameras and two timestamps per variant (88 RGB images). Room geometry hashes are invariant across each family's camera, material, detail, illumination and exposure interventions. People are disabled in this sweep. The receipt binds the actual native executable, source identity, target, feature flags and locked package versions. All cases are retained, including failures; no model inference filters pairs. The project page provides the complete matched annotation gallery and machine-readable receipt.

Captures export a full row-major pixel-space $K$, image dimensions, half-pixel centers, a versioned centered pinhole lens contract, normalized playback progress, eased trajectory progress and optional explicitly assigned seconds. Unassigned physical time remains unknown. Short handheld paths sample local metric translations and yaw/pitch/roll increments independently of a moving look-target, retaining swept collision checks. Requested high/low/zero proxy-overlap strata are selected before placement retries. Their all-ray proxy estimates are stored separately from rendered directed overlap over valid source pixels. A proxy-negative pair can retain a small rendered overlap. Lens distortion and sampled principal points are not implemented.

Table 6 reports means over directed rendered pairs, and the maximum per-view 99th-percentile self-reprojection error. There are 10 exact duplicate RGB images: the baseline/overlap interventions deliberately reuse the reference camera. These counts measure exact repetition in this matched study, not perceptual redundancy across a training population. Sensor white balance, blur, noise and optional JPEG are separate seeded RGB-only export transforms with individual operator regressions; they are not applied in this render sweep. This bounded check does not establish photographic realism, a calibrated sensor model, a fitted training mixture or downstream transfer.

| Factor | Shared pixels (%) | Baseline (m) | Reproj. p99 max (px) |
|---|---|---|---|
| reference | 45.5688 | 3.559 | 0.00010 |
| portrait | 50.4379 | 1.736 | 0.00018 |
| wide baseline | 35.7425 | 5.603 | 0.00010 |
| handheld | 53.3032 | 2.050 | 0.00008 |
| low overlap | 11.9214 | 2.235 | 0.00010 |
| no proxy overlap | 0.0000 | 3.001 | 0.00006 |
| material seed | 45.5688 | 3.559 | 0.00010 |
| low detail | 45.5688 | 3.559 | 0.00010 |
| lighting seed | 45.5688 | 3.559 | 0.00010 |
| dim illumination | 45.5688 | 3.559 | 0.00010 |
| exposure | 45.5688 | 3.559 | 0.00010 |

Table 6: Prespecified factor checks on matched room geometry. Directed overlap uses valid source pixels; the zero target refers to placement proxies.

### 5.5 Population distributions and publication contract

Room and object counts retain zero-instance rooms. Camera and photometric histograms state their own sample denominators. Placement heatmaps use 33 path samples per camera and normalize each panel independently. Feature shortcuts do not change population denominators. A seed's bit width and continuous parameter ranges do not establish ten-million-sample diversity; correlations, placement constraints and rendering affect distinguishable outcomes. Frozen SigLIP2 embeddings support separate redundancy audits, but no embedding-rank or downstream-learning result is claimed here.

The project page, paper, figures and calibration archives are built by a single Rust publication pipeline from completed capture manifests. All displayed cohorts must match the current compiled generator identity, which binds crate version, geometry grammar and source/dependency hashes. The publisher validates scene/camera/time identities, every requested channel, exact membership masks, annotation checks and population denominators before staging artifacts. The release/deployment gate checks input and output hashes and rejects historical cohorts. Author-written method text and measurements have distinct roles; publication attestation does not imply photographic or downstream validation.

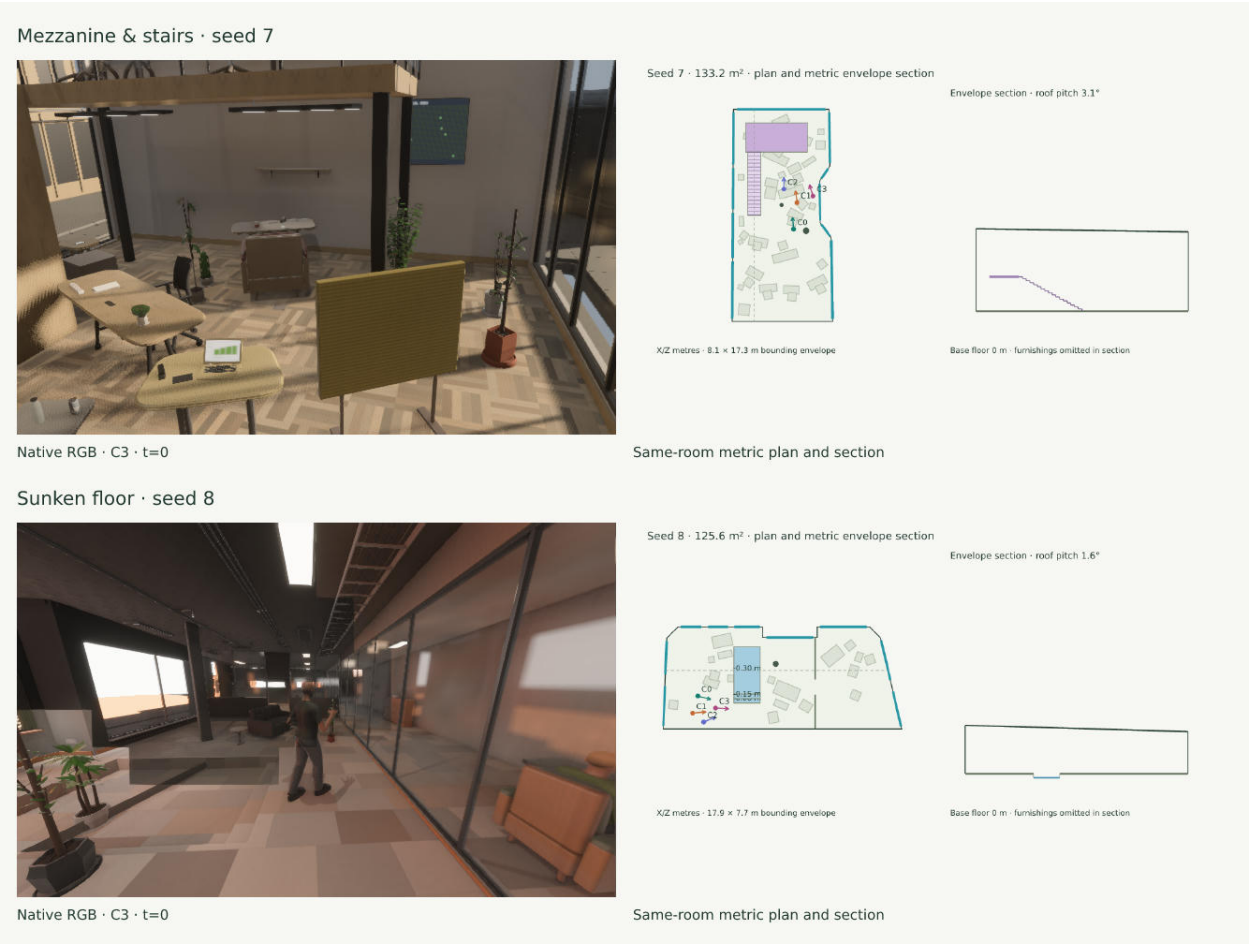



Figure 4: Native RGB and metric plans of structurally selected rooms. Raised and depressed levels, decks and stairs are generated geometry. Cyan marks glazing; orange/blue denote raised/sunken floors; violet denotes mezzanines and stairs.

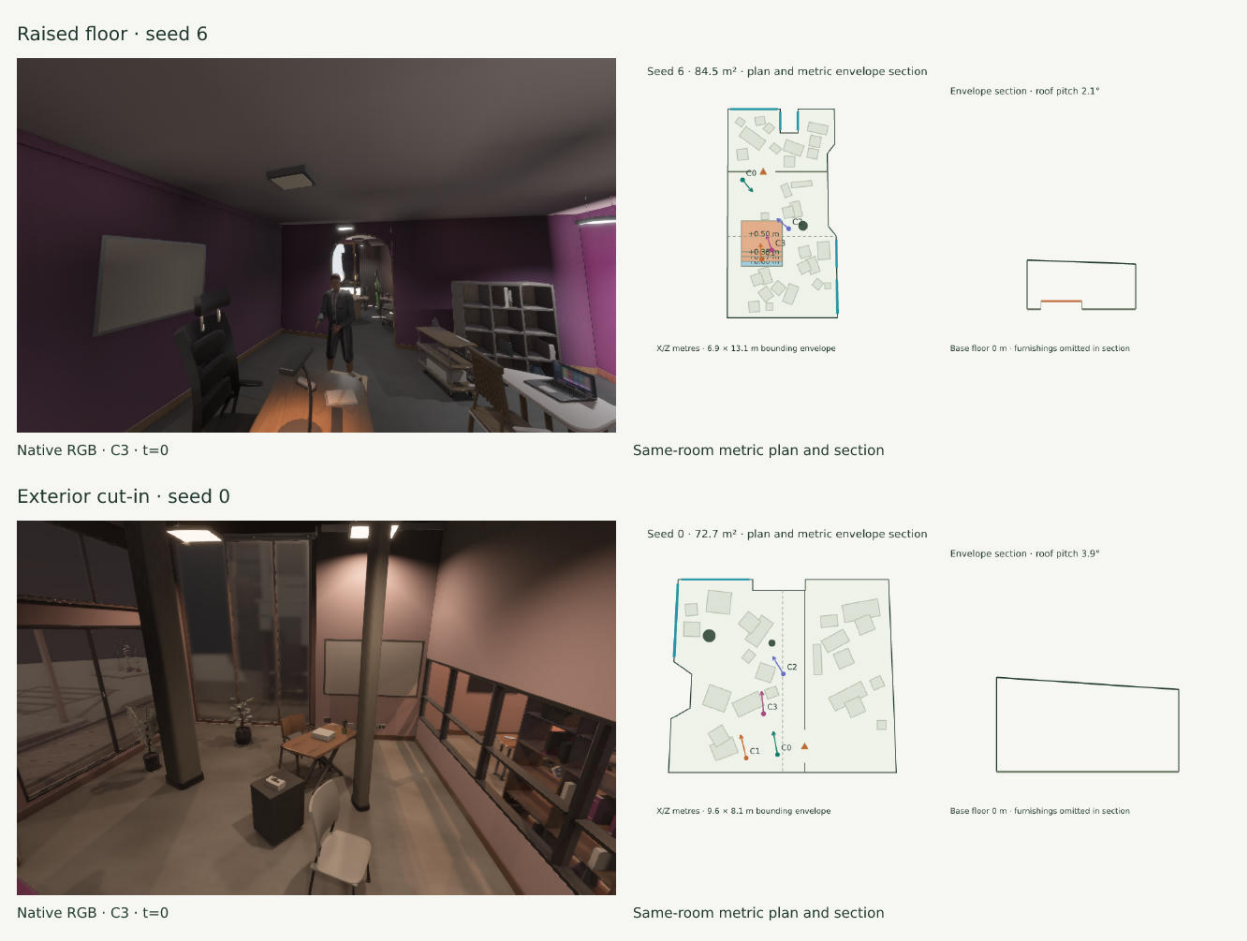



Figure 5: Native RGB paired with the corresponding envelope programs. Floor levels, portals, polygonal outlines and apertures vary; features can be outside a particular camera view.




Figure 6: Twelve metric envelope programs selected by architectural feature coverage from the full consecutive population. Plans are diagrams, not twelve additional RGB renders.

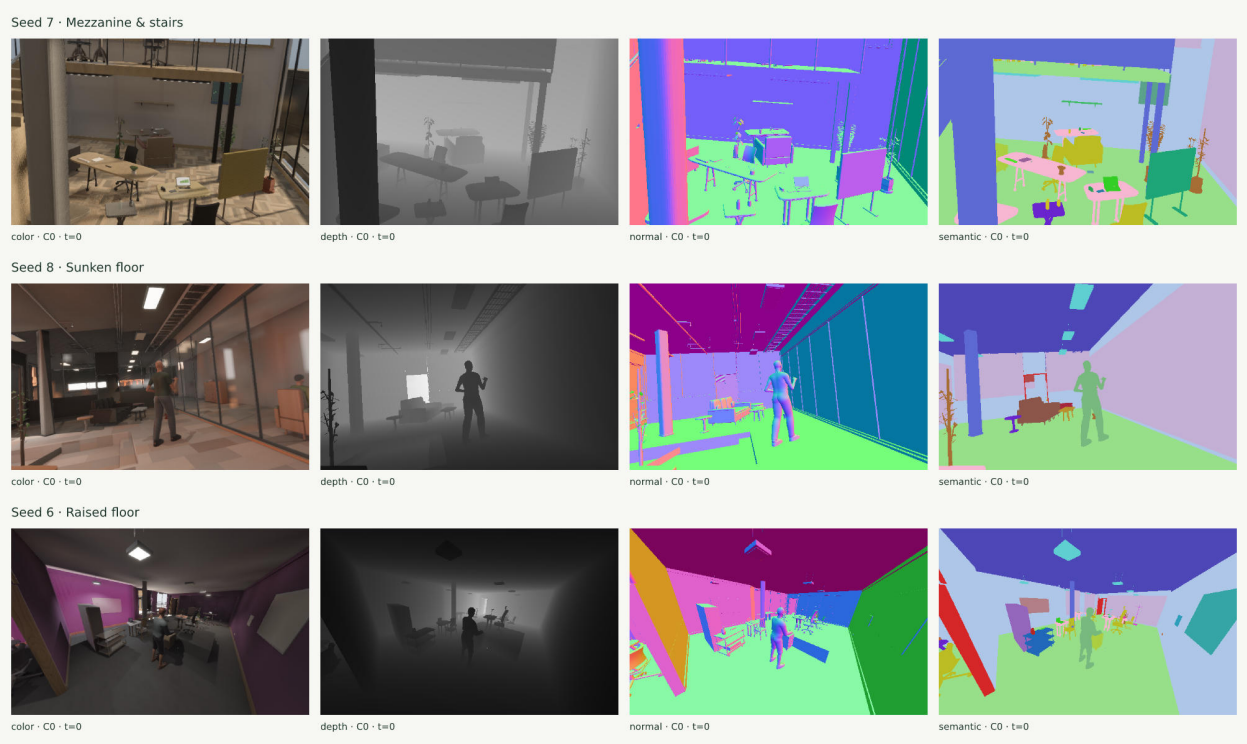



Figure 7: Matched RGB, depth, view-space normals and semantic display previews, camera 0 at t=0. All six annotations, camera identities and endpoint times remain accessible in the explorer.

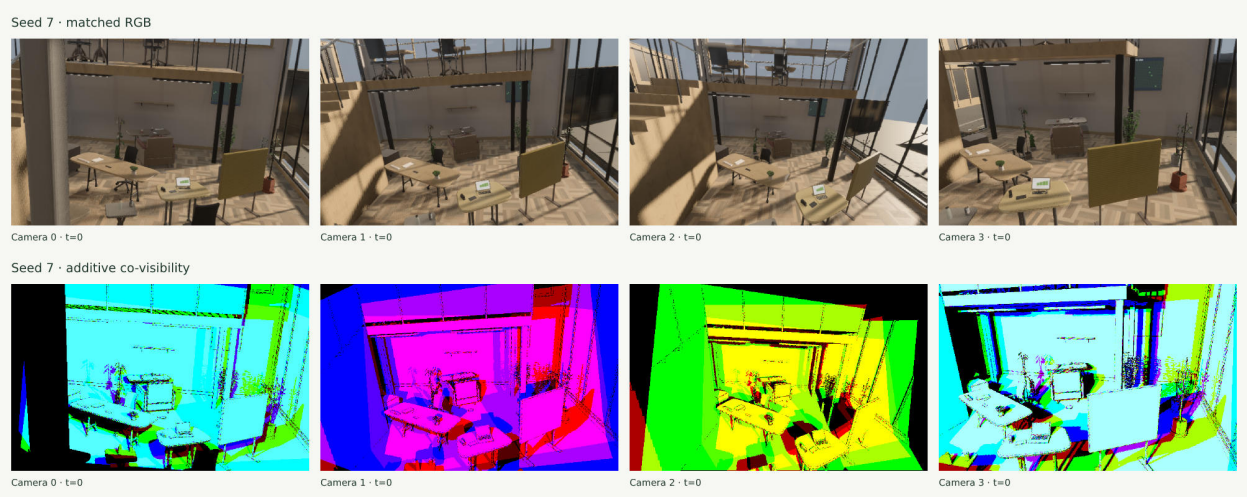



Figure 8: Seed 7 at t=0: four native RGB views above their production co-visibility annotations. Columns retain the same camera/time identity. All captured rooms expose this annotation in the same explorer.

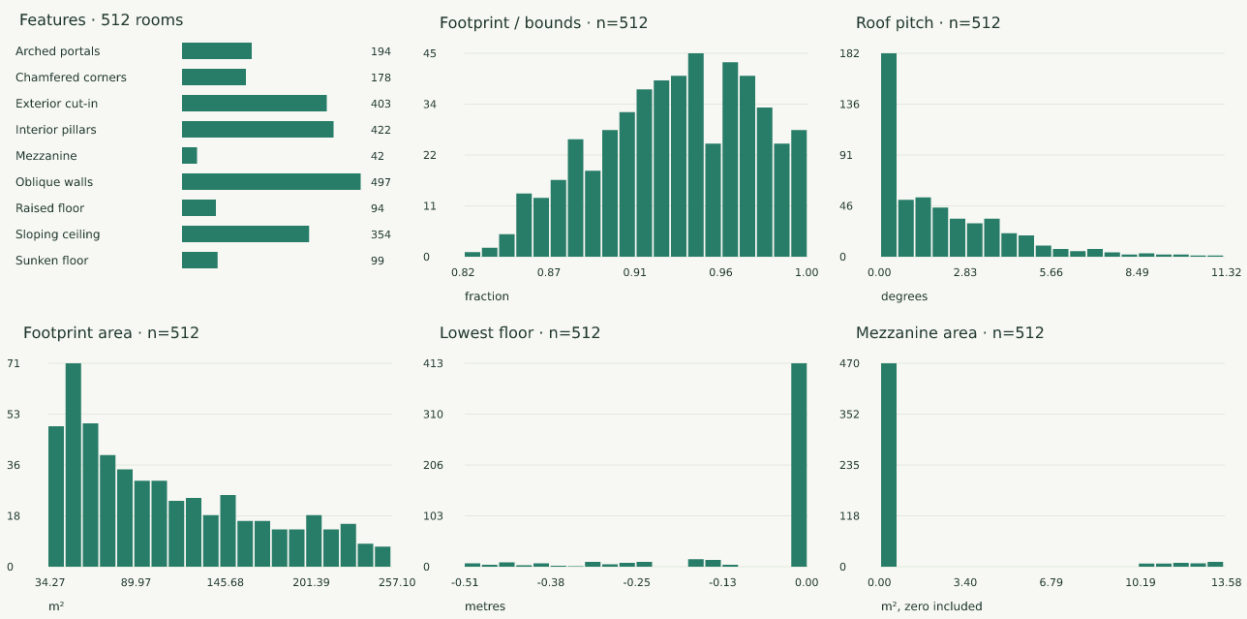



Figure 9: Architectural feature counts and continuous parameter distributions from the complete population; absent features are retained.

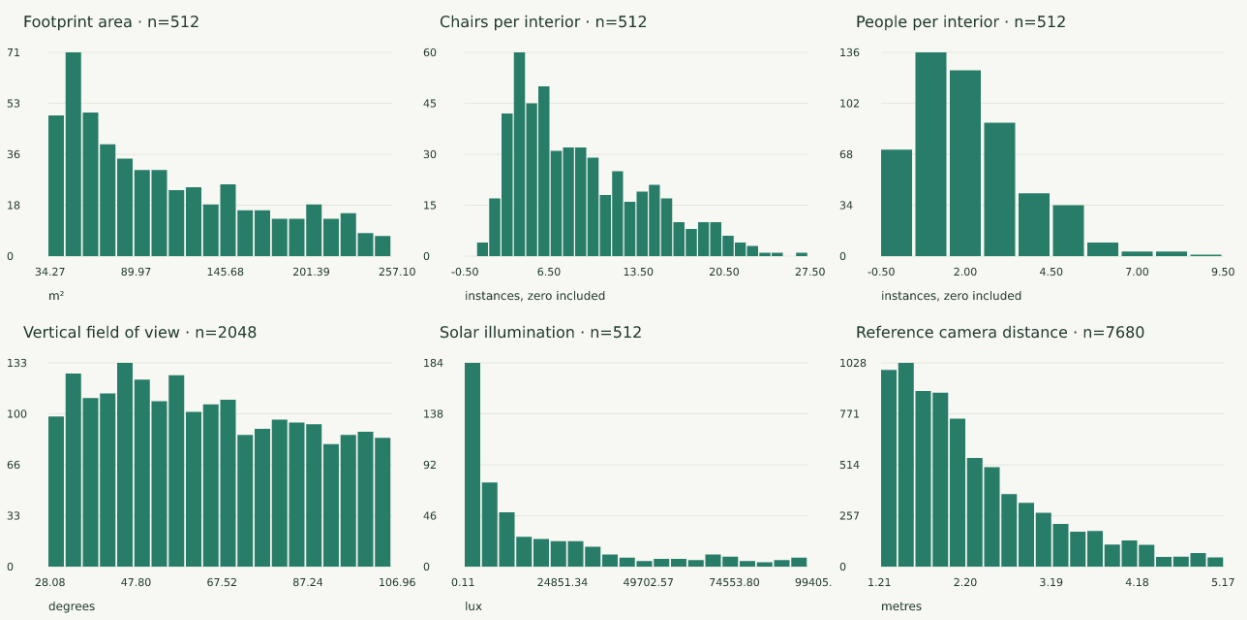



Figure 10: Furnishing, lighting and camera variation in the same population. Every panel states its denominator.

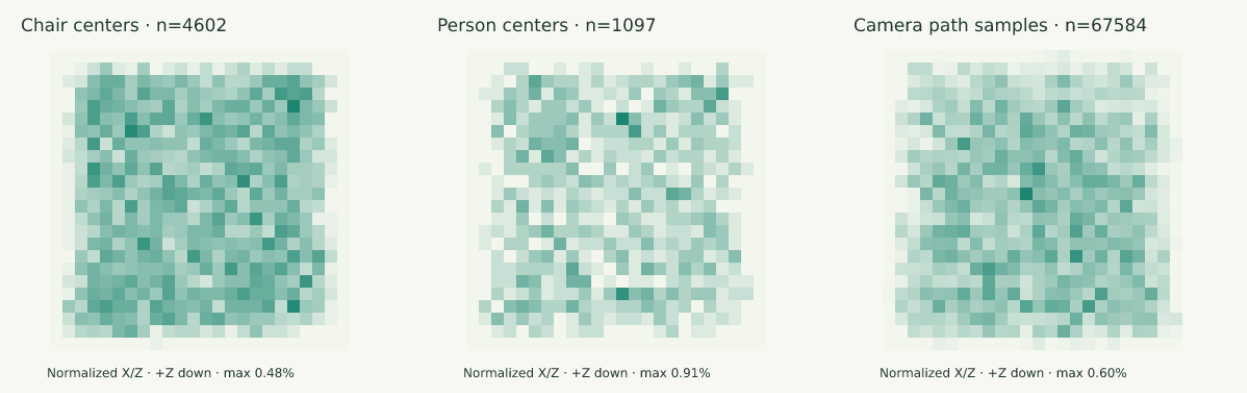



Figure 11: Chair centers, human centers and camera path samples in normalized room X/Z coordinates, with +Z down. Each panel reports its own sample count and color scale.

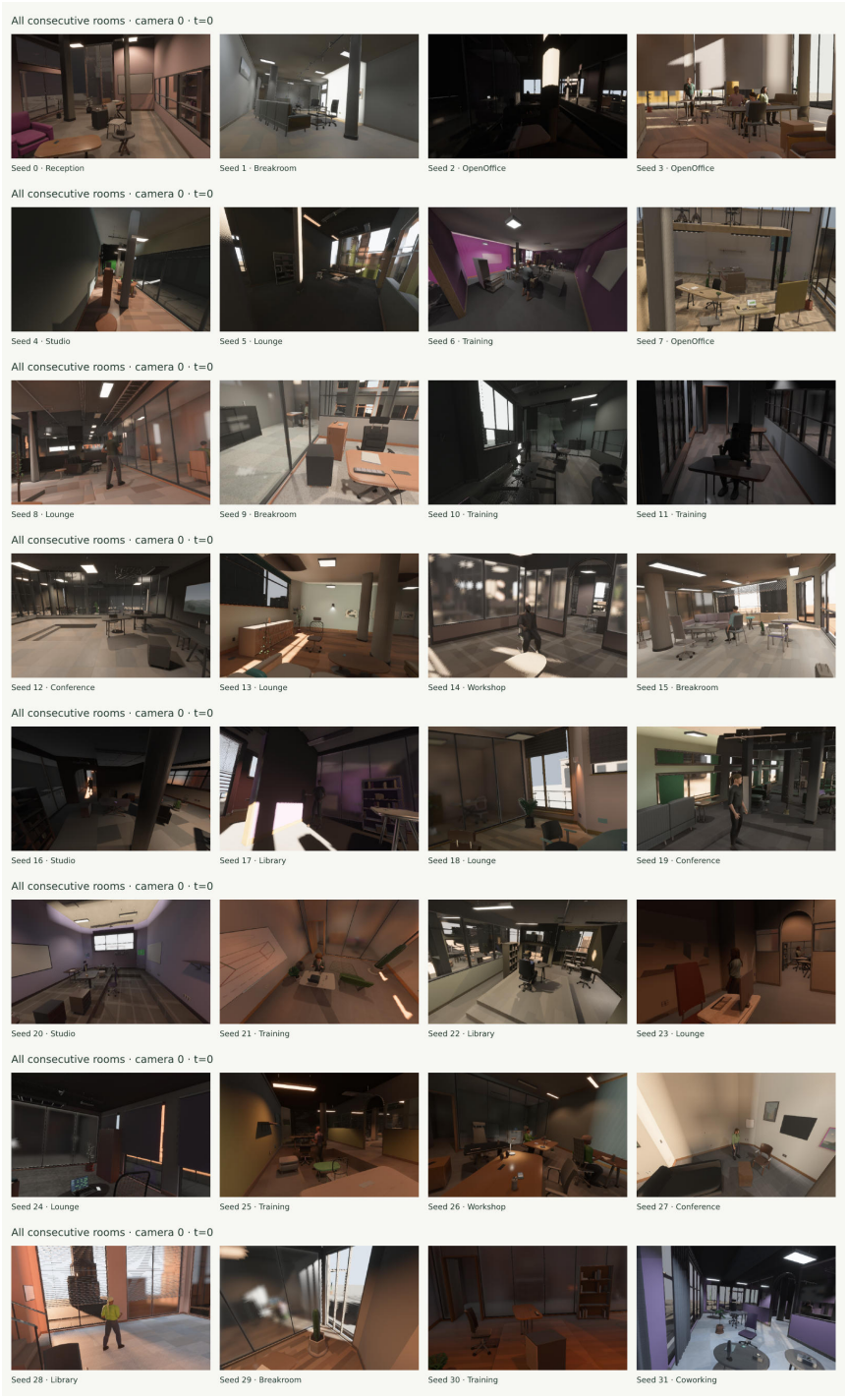



Figure 12: Every consecutively rendered room, camera 0 at t=0, including dark and sparsely furnished rooms. All four cameras, endpoints and six modes remain in the project-page explorer.

## 6 Camera-count and baseline audit

The current sampler was evaluated on 64 consecutive matched rooms (seeds 0–63), 1–12 cameras and baseline values 0, 0.5 and 1 at aspect ratio 1.60. Geometry, people and appearance were unchanged between requests; complete camera sets were resampled, so different counts are not necessarily nested prefixes. 0 of 2304 requested sets were rejected by the bounded sampler; all failures remain in the downloadable report. Camera-center and optical-axis measurements use 33 synchronized path samples. Reference-edge overlap uses the production $13\times9$ first-surface proxy at five times, rather than rendered pixels.

At one camera the baseline imposes no pair constraint. Increasing camera count does not increase the minimum nonreference pair separation: the same $0.04 + 0.46b$ metre threshold applies. Group covariance spread constrains overall anisotropy, not every nearest neighbor. Visible-region target variation samples reference-visible surface patches and ranks feasible candidates for count-aware nearest-peer spacing, optical-axis separation and additional primary-room proxy surface cells. The soft distance scale is room diagonal divided by the square root of camera count. Temporal paths can remain related deformations of an existing route; varied gaze is not independent route topology. There is no explicit 90-degree optical-angle cap.

| $b$ | Cameras | Accepted | NN (m) | Max angle (°) | Targets (m) |
|---|---|---|---|---|---|
| 0.0 | 2 | 64/64 | 0.16 | 1.5 | 0.02 |
| 0.0 | 4 | 64/64 | 0.13 | 3.3 | 0.04 |
| 0.0 | 8 | 64/64 | 0.11 | 5.4 | 0.06 |
| 0.0 | 12 | 64/64 | 0.09 | 6.1 | 0.06 |
| 0.5 | 2 | 64/64 | 2.19 | 23.0 | 0.19 |
| 0.5 | 4 | 64/64 | 1.46 | 52.8 | 1.30 |
| 0.5 | 8 | 64/64 | 1.03 | 71.0 | 1.96 |
| 0.5 | 12 | 64/64 | 0.79 | 87.1 | 2.26 |
| 1.0 | 2 | 64/64 | 3.58 | 62.9 | 1.25 |
| 1.0 | 4 | 64/64 | 2.47 | 109.2 | 3.10 |
| 1.0 | 8 | 64/64 | 1.24 | 142.4 | 4.96 |
| 1.0 | 12 | 64/64 | 0.98 | 159.1 | 5.29 |

Table 7: Current matched placement audit. NN is the pooled camera-level initial nearest-neighbor median; angle and target diameter are room-level medians. Failed sets have no placement measurements. These denominators differ from rendered pixel overlap.

Similar absolute trajectories have RMS center separation below 0.15 m and RMS optical-axis difference below $5^\circ$. Shape similarity instead removes each start position and normalizes the RMS displacement residual by the larger RMS travel, with a diagnostic threshold of 0.25; static pairs are excluded. Close stereo views can intentionally satisfy the pose threshold. A common-axis least-squares residual measures convergence of initial optical-axis lines; parallel or ill-conditioned lines have no reported focus. Neither statistic proves perceptual repetition or reconstruction utility.




Figure 13: Camera-count distributions at three baseline settings. Room-level viewing-angle/target statistics and camera-level spacing use their own denominators. Repeated path shapes are measured separately from repeated absolute poses.

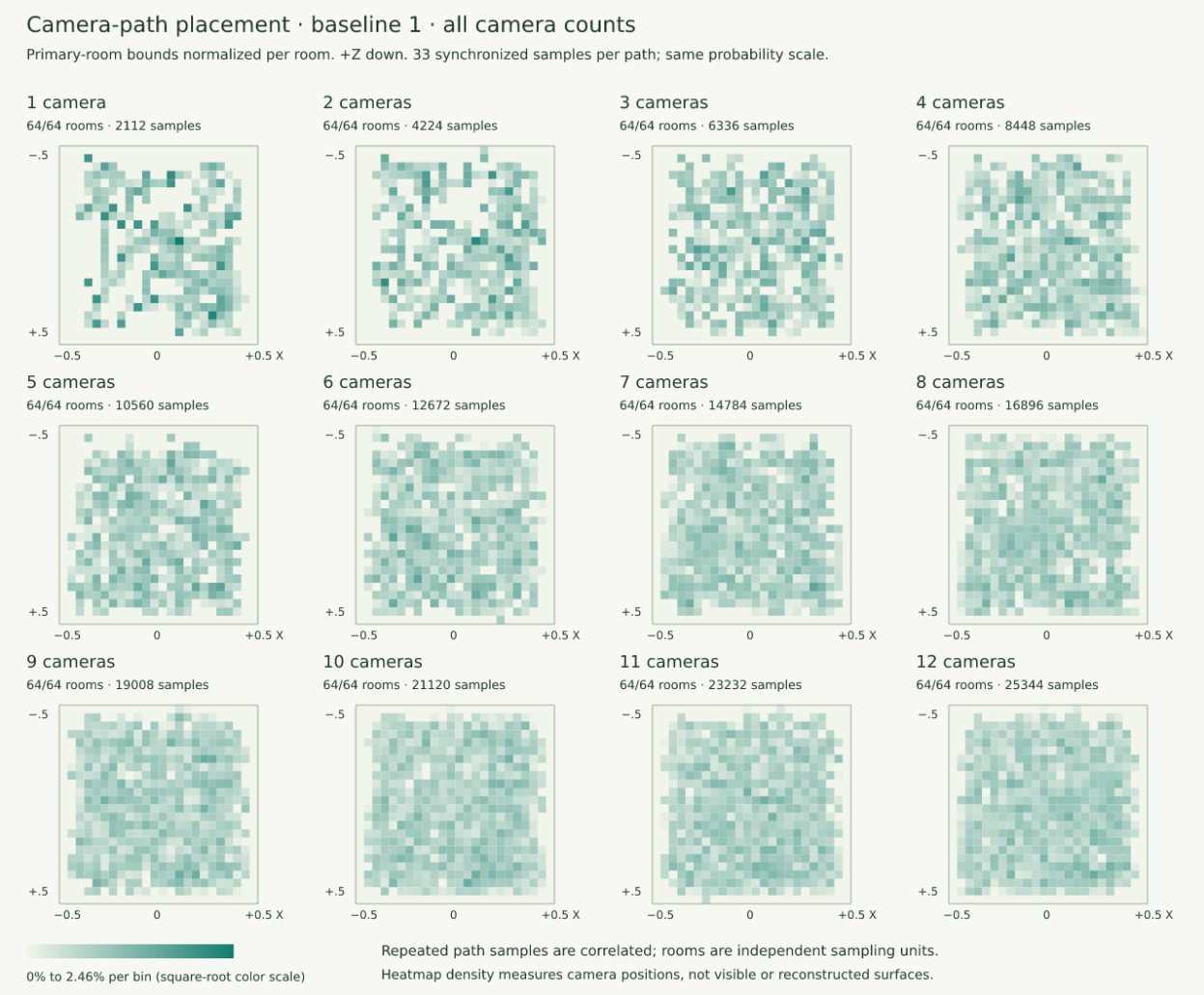



Figure 14: Baseline 1 camera-path heatmaps for 1–12 views: 33 synchronized samples per camera, primary-room bounds normalized per room and +Z down. A common probability color scale is used. These show positions, not reconstructed-surface coverage.

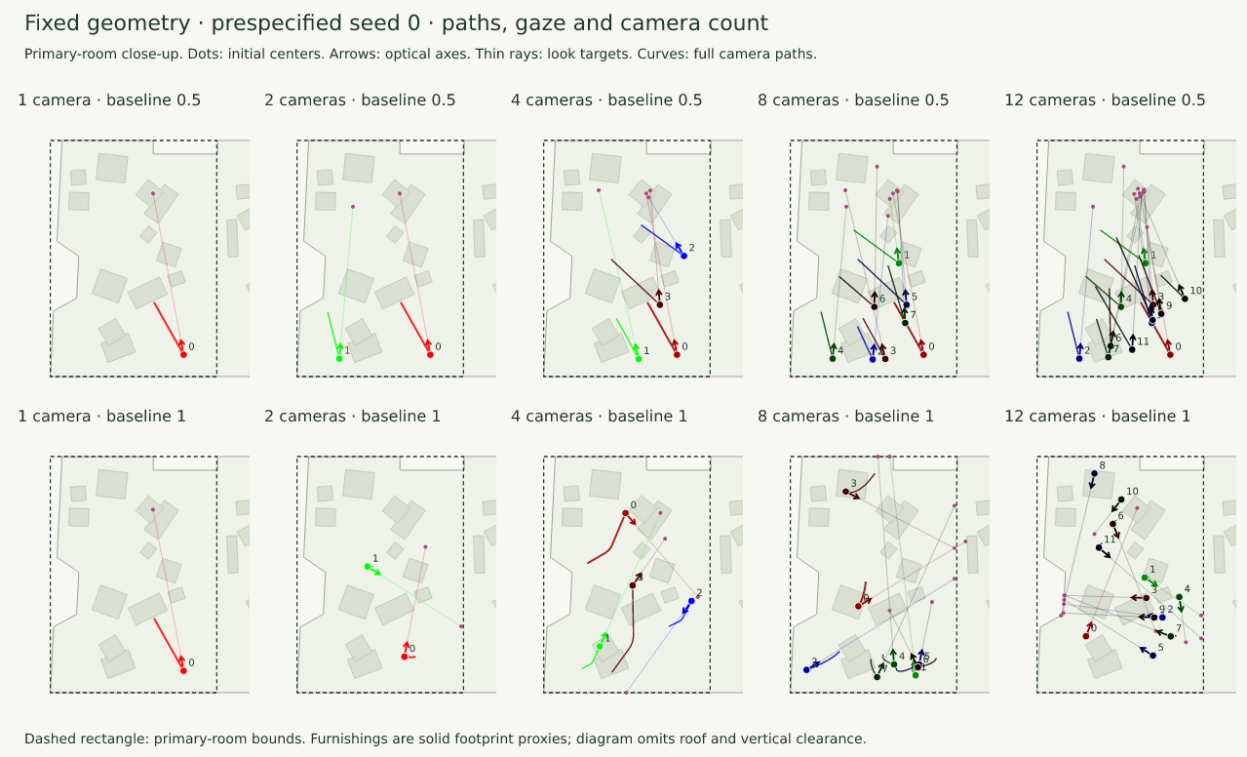



Figure 15: Prespecified first room, fixed geometry and people, default/wide baselines and increasing camera counts. Full paths, initial optical directions and look targets expose the shared-content proposal. Rejected requests are retained.

### 6.1 Native rendered overlap across camera counts

The rendered check uses 2 prespecified matched room families, 12 configurations and 312 camera/time views: counts 2, 4, 8 and 12 at all three baselines and both trajectory endpoints, with 320×200 images, static human density 0.25 and native shadows, material synthesis and diffuse GI retained. The executable and every channel are hash-bound. Rendered centers and optical axes match the CPU audit within $10^{-4}$ m and $10^{-5}$ per direction component. Table 8 separates mean directed pair overlap from mean source-view visibility in any peer, and reports the smallest reference-edge overlap over the sampled rooms/times. These do not measure unique scene-surface coverage.

| $b$ | Cameras | Pair (%) | Any peer (%) | Min ref. (%) | Components |
|---|---|---|---|---|---|
| 0 | 2 | 84.5 | 84.5 | 70.9 | 1 |
| 0 | 4 | 84.2 | 94.0 | 70.9 | 1 |
| 0 | 8 | 83.5 | 97.7 | 69.3 | 1 |
| 0 | 12 | 83.8 | 98.8 | 67.1 | 1 |
| 0.5 | 2 | 51.7 | 51.7 | 37.7 | 1 |
| 0.5 | 4 | 54.3 | 81.3 | 36.1 | 1 |
| 0.5 | 8 | 57.1 | 91.0 | 33.7 | 1 |
| 0.5 | 12 | 55.5 | 93.8 | 37.1 | 1 |
| 1 | 2 | 53.9 | 53.9 | 42.0 | 1 |
| 1 | 4 | 41.0 | 73.6 | 16.6 | 1 |
| 1 | 8 | 41.2 | 87.9 | 16.1 | 1 |
| 1 | 12 | 32.7 | 94.9 | 16.4 | 1 |

Table 8: Native first-surface co-visibility; glass follows the solid annotation convention. Components is the largest number of connected components when an undirected edge requires at least 5% shared pixels in both directions. The two room families are the independent units; camera pairs and endpoints are correlated.

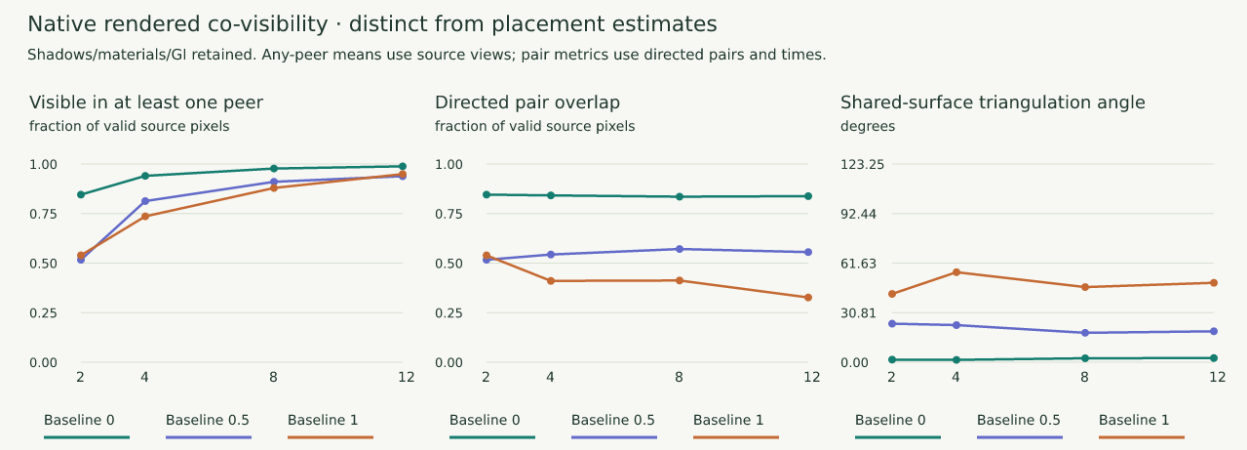



Figure 16: Measured native visibility and triangulation at increasing camera counts. Any-peer visibility is averaged over source views, other means over directed pairs. More peer views increase any-peer visibility without proving broader unique surface coverage.

### 6.2 Visual effect of camera baseline

Each column resamples a complete four-camera group over identical geometry, people, materials and lighting. Camera indices identify group members, rather than fixed physical viewpoints across columns. These are the first two prespecified native families, with every view retained at t=0 and no per-image exposure correction. They contribute no extra rooms to the population denominator.

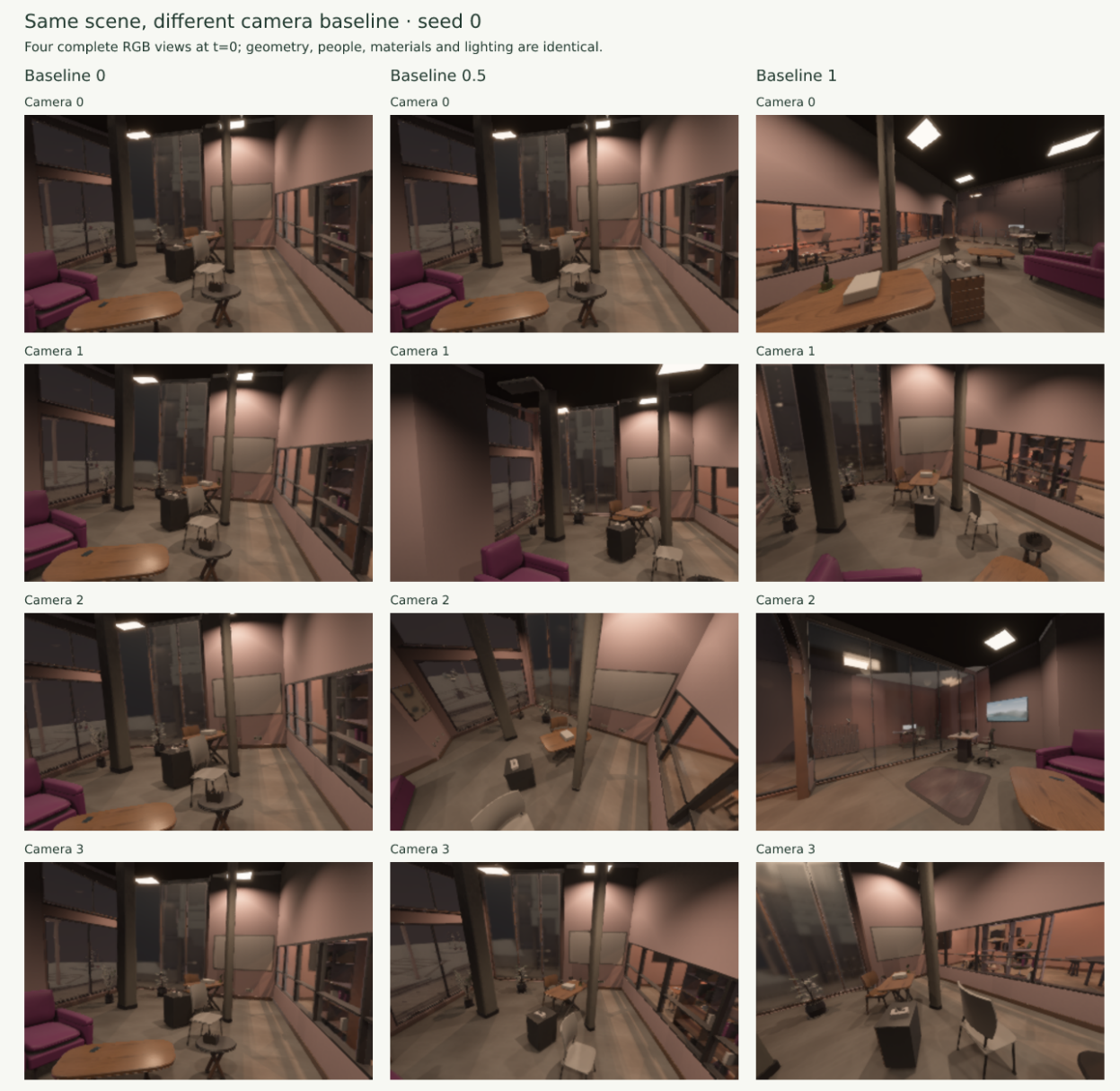



Figure 17: Fixed room 0: narrow, default and wide baseline settings (0, 0.5 and 1) from left to right. Rows retain all four native RGB views at 320×200. Perspective and shared content change with the camera policy; scene content is invariant.




Figure 18: Fixed room 1: narrow, default and wide baseline settings (0, 0.5 and 1) from left to right. Rows retain all four native RGB views at 320×200. Perspective and shared content change with the camera policy; scene content is invariant.

# 7 Object geometry diversity

## 7.1 Fixed-envelope protocol

We evaluate all 64 furnishing/prop categories on 256 consecutive construction seeds each: 16384 triangle assemblies. Each category holds its metric placement envelope, orientation and finish seed fixed. This isolates internal construction from changes in object size, pose or color; it deliberately does not measure the full joint scene distribution. The cable fixture varies a controlled centerline; actual endpoint connectivity and collision are separate room-level checks. Humans and architecture are evaluated elsewhere.

For each mesh, six orthographic first-surface depth maps sample pixel centers at $24 \times 24$ resolution per direction. Depth is normalized by the fixed envelope extent. Pair distance is mean absolute depth difference over the union of hit pixels; missing-versus-present surface contributes one. Greedy coverage retains a sample only when its distance to every earlier representative exceeds the threshold. We report thresholds 0.005, 0.015 and 0.030, and retain seed order. These are finite-resolution, order-dependent coverage counts, not numbers of independent geometries, perceptual embedding volume, or a proof of a multiplicative increase in dataset utility. Thin features and content on nearly planar surfaces can be under-resolved; saturated categories need more samples or a finer descriptor.

All audited assemblies passed finite position/normal/UV, unit-normal, winding, nondegenerate-triangle, index, semantic-label and reserved-envelope checks (maximum allowed overrun 3 mm). This does not prove watertightness, lack of every self-intersection, or physical plausibility. Separate regression tests cover support, cable attachment, overlapping planar faces and deterministic replay. Mesh build time excludes placement, materials, uploads and rendering and is not completed-room throughput.

Six examples per category are selected by deterministic farthest-spread sampling from the same depth descriptors, beginning with seed zero. The native renderer reuses one device, with fixed cameras, exposure, lights and finish seed within each category. The project-page atlas provides RGB and normal views, every example's program inputs, all sampled parameter marginals and the complete machine-readable measurements. These selected examples are correlated draws, not one-parameter interventions; inactive conditional controls are not evidence of realized shape diversity.

## 7.2 Category measurements

Table 9: Current construction coverage at three distances; triangles and CPU build milliseconds are per-object medians. All categories use 256 seeds.

| Category | $C_{.005}$ | $C_{.015}$ | $C_{.030}$ | Triangles | ms |
|---|---|---|---|---|---|
| Table | 255 | 246 | 225 | 2272 | 0.147 |
| Desk | 255 | 246 | 219 | 2184 | 0.125 |
| Chair | 255 | 252 | 241 | 4444 | 0.148 |
| Sofa | 256 | 252 | 176 | 11552 | 0.371 |
| CoffeeTable | 256 | 255 | 236 | 2200 | 0.058 |
| Cabinet | 46 | 20 | 7 | 1744 | 0.045 |
| Bookcase | 256 | 251 | 163 | 9576 | 0.273 |
| Plant | 256 | 256 | 256 | 5240 | 0.229 |
| TrashCan | 219 | 138 | 54 | 1452 | 0.031 |
| Whiteboard | 51 | 26 | 9 | 646 | 0.035 |
| Display | 91 | 47 | 34 | 132 | 0.006 |
| Laptop | 252 | 200 | 122 | 8012 | 0.212 |
| Monitor | 256 | 255 | 250 | 668 | 0.024 |
| Notebook | 3 | 2 | 1 | 540 | 0.019 |
| Mug | 212 | 97 | 36 | 1284 | 0.034 |
| Books | 256 | 232 | 137 | 2160 | 0.087 |
| Rug | 167 | 66 | 30 | 384 | 0.017 |
| WallArt | 15 | 3 | 2 | 542 | 0.025 |
| Clock | 159 | 58 | 9 | 5560 | 0.234 |
| FloorLamp | 256 | 256 | 256 | 2416 | 0.078 |
| Keyboard | 252 | 235 | 150 | 7020 | 0.179 |
| Mouse | 251 | 210 | 115 | 1040 | 0.056 |
| WaterBottle | 160 | 67 | 22 | 3104 | 0.095 |
| PenHolder | 256 | 256 | 256 | 752 | 0.033 |
| Printer | 171 | 49 | 15 | 776 | 0.037 |

| Category | $C_{.005}$ | $C_{.015}$ | $C_{.030}$ | Triangles | ms |
|---|---|---|---|---|---|
| StorageBox | 188 | 55 | 18 | 452 | 0.035 |
| CoatRack | 233 | 205 | 159 | 3280 | 0.122 |
| Bag | 246 | 120 | 33 | 736 | 0.041 |
| CoffeeCup | 112 | 65 | 30 | 956 | 0.032 |
| SodaCan | 11 | 2 | 1 | 1752 | 0.050 |
| Notepad | 43 | 23 | 10 | 240 | 0.010 |
| Pencil | 59 | 21 | 9 | 116 | 0.007 |
| Microphone | 168 | 152 | 103 | 1076 | 0.029 |
| Phone | 66 | 20 | 13 | 200 | 0.009 |
| WallOutlet | 14 | 7 | 3 | 1252 | 0.041 |
| LightSwitch | 31 | 14 | 6 | 724 | 0.027 |
| SideTable | 256 | 250 | 236 | 2200 | 0.081 |
| MobileCart | 147 | 92 | 56 | 3348 | 0.093 |
| DrawerUnit | 13 | 3 | 3 | 1400 | 0.043 |
| Bench | 23 | 13 | 8 | 1136 | 0.035 |
| Ottoman | 144 | 73 | 32 | 1840 | 0.057 |
| RoomDivider | 23 | 17 | 17 | 540 | 0.049 |
| DeskLamp | 233 | 206 | 164 | 896 | 0.028 |
| CoffeeMaker | 130 | 38 | 21 | 1024 | 0.040 |
| Kettle | 150 | 39 | 14 | 1628 | 0.046 |
| Speaker | 197 | 84 | 27 | 1232 | 0.059 |
| Headphones | 250 | 213 | 136 | 2624 | 0.083 |
| Stapler | 152 | 52 | 18 | 404 | 0.015 |
| TapeDispenser | 201 | 105 | 44 | 648 | 0.023 |
| DocumentTray | 34 | 23 | 12 | 2052 | 0.045 |
| DeskOrganizer | 8 | 8 | 8 | 888 | 0.027 |
| TissueBox | 256 | 201 | 61 | 1292 | 0.032 |
| Vase | 252 | 241 | 183 | 384 | 0.017 |
| Bowl | 218 | 141 | 68 | 384 | 0.013 |
| DeskFan | 71 | 35 | 18 | 4276 | 0.133 |
| Projector | 93 | 10 | 2 | 2728 | 0.075 |
| ProjectionScreen | 18 | 10 | 8 | 514 | 0.015 |
| PowerStrip | 4 | 2 | 1 | 2052 | 0.088 |
| Cable | 209 | 188 | 143 | 496 | 0.016 |
| WallShelf | 34 | 30 | 17 | 700 | 0.036 |
| NoticeBoard | 1 | 1 | 1 | 6600 | 0.226 |
| WallSconce | 235 | 174 | 103 | 344 | 0.023 |
| CeilingPendant | 108 | 38 | 16 | 464 | 0.020 |
| FloorOutlet | 73 | 72 | 31 | 668 | 0.025 |




Figure 19: Matched studio construction examples: Chair, Table, Sofa, Plant. All columns within each row share dimensions, viewpoint, lights and finish seed. Seed labels link to the full sampled parameter grid in the downloadable atlas. Example selection emphasizes shape spread, not frequency or image quality; the table includes every sampled seed.




Figure 20: Matched studio construction examples: Keyboard, Mouse, FloorLamp, CoatRack. All columns within each row share dimensions, viewpoint, lights and finish seed. Seed labels link to the full sampled parameter grid in the downloadable atlas. Example selection emphasizes shape spread, not frequency or image quality; the table includes every sampled seed.




Figure 21: Matched studio construction examples: StorageBox, Printer, Bag, Headphones. All columns within each row share dimensions, viewpoint, lights and finish seed. Seed labels link to the full sampled parameter grid in the downloadable atlas. Example selection emphasizes shape spread, not frequency or image quality; the table includes every sampled seed.

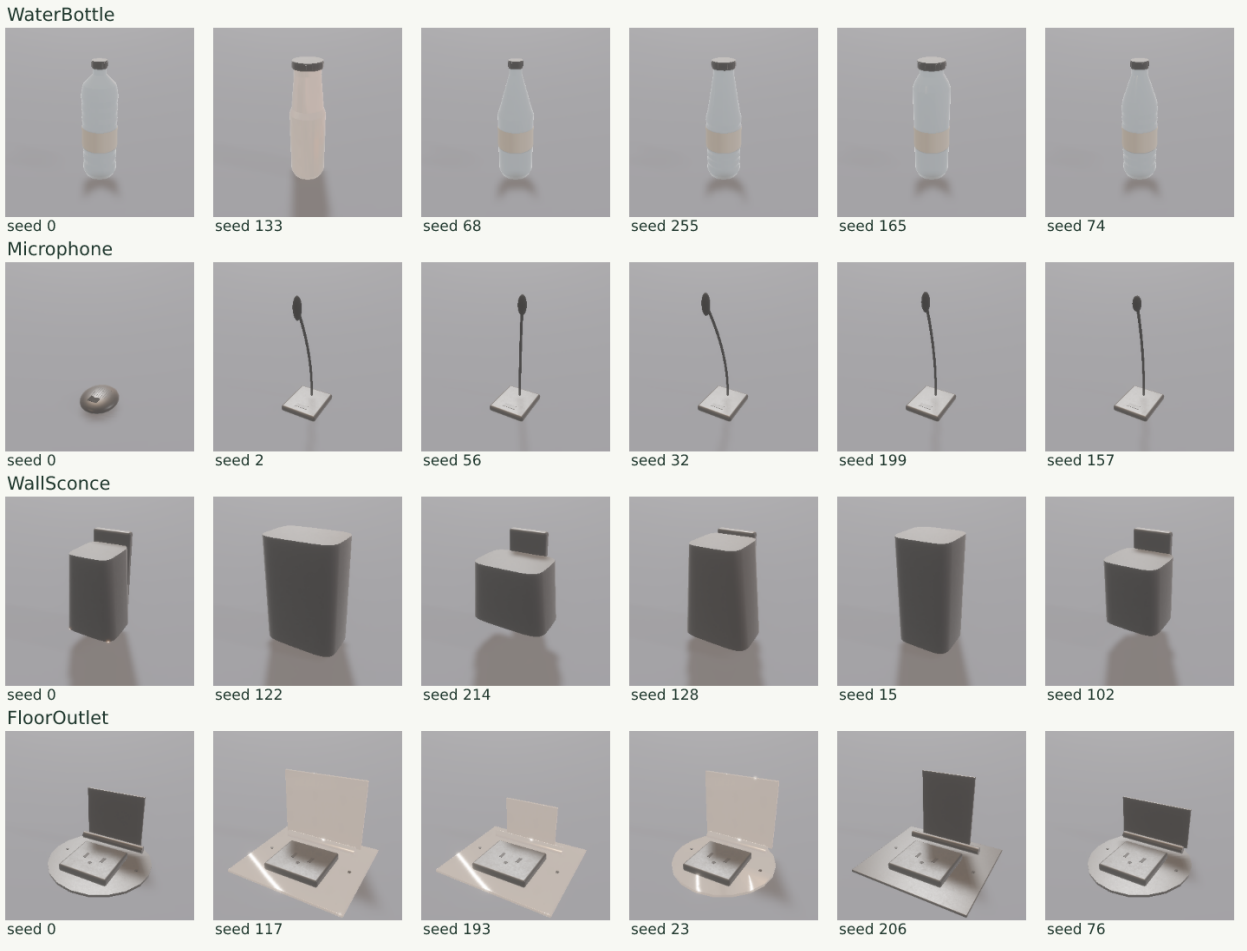



Figure 22: Matched studio construction examples: WaterBottle, Microphone, WallSconce, FloorOutlet. All columns within each row share dimensions, viewpoint, lights and finish seed. Seed labels link to the full sampled parameter grid in the downloadable atlas. Example selection emphasizes shape spread, not frequency or image quality; the table includes every sampled seed.




| Construction seed | 0 | 228 | 234 | 253 | 179 | 120 |
|---|---|---|---|---|---|---|
| Shade height (m) | 0.248 | 0.235 | 0.201 | 0.239 | 0.332 | 0.348 |
| Radius / envelope width | 0.402 | 0.327 | 0.357 | 0.414 | 0.318 | 0.349 |
| Upper / lower radius | 0.993 | 0.630 | 0.716 | 0.674 | 0.932 | 0.510 |
| Stem offset / width | 0.053 | 0.031 | -0.087 | 0.096 | 0.008 | -0.052 |
| Shade sides | 32 | 32 | 32 | 32 | 32 | 32 |
| Support code | 2 | 0 | 1 | 1 | 1 | 2 |

Figure 23: A construction parameter grid with values taken from the pictured meshes. Each lamp occupies the same $0.55 \times 1.65 \times 0.55$ m reserved envelope. Support code 0 is a disk and stem, 1 a wooden tripod, and 2 four metal legs. A 32-sided shade approximates a circular section; smaller counts produce polygonal shades. Stem offset and shade radius are jointly bounded by the reserved envelope. The bulb and light emitter use the same computed position. This grid shows correlated sampled programs, not isolated causal effects.

### 7.3 Sampled controls and their interpretation

The following inventory lists the sampled construction inputs recorded by each builder. Some controls are conditional on a construction choice (for example, a phone's rear lenses affect only face-down configurations); a range does not imply independent influence. Observed ranges, eight-bin numeric marginals, categorical frequencies and all seed-level values accompany the project-page grids. A small coverage count for a flat notebook, notice board or outlet is retained explicitly rather than counting different textures as different geometry.

| Category | Recorded controls |
|---|---|
| Table | capsule_ blend, edge_ bevel, frame_ bend_ m, leg_ aspect, leg_ inset, leg_ radius, leg_ rake, leg_ taper, outline_ exponent, pedestal_ base_ aspect, pedestal_ base_ radius, pedestal_ radius, spider_ sweep, support, taper, top_ surface, top_ thickness, trestle_ spread |
| Desk | capsule_ blend, edge_ bevel, frame_ bend_ m, leg_ aspect, leg_ inset, leg_ radius, leg_ rake, leg_ taper, outline_ exponent, pedestal_ base_ aspect, pedestal_ base_ radius, pedestal_ radius, spider_ sweep, support, taper, top_ surface, top_ thickness, trestle_ spread |
| Chair | arm_ height, arm_ pad_ length, arm_ pad_ roundness, arm_ pad_ width_ m, arm_ support, armrests, back_ construction, back_ width_ fraction, base_ radius_ fraction, curvature, frame_ bend_ m, headrest, headrest_ curvature_ m, headrest_ height_ m, headrest_ width_ fraction, leg_ splay, lumbar, mesh_ pitch_ m, recline, seat_ crown, seat_ depth_ fraction, seat_ dish_ m, seat_ roundness, seat_ width_ fraction, shell_ thickness, shoulder_ flare, spindle_ pitch, spoke_ count, taper |
| Sofa | arm_ crown, arm_ rise_ m, arm_ roundness, arm_ width, back_ tilt, chaise, cushion, cushion_ crown, cushion_ dish_ m, cushion_ gap_ m, exposed_ frame, left_ return, leg_ height, module_ asymmetry, modules, pillow_ roll, pillow_ width_ m, pillows, roundness, seam_ radius_ m, seat_ height, upholstery |
| CoffeeTable | capsule_ blend, edge_ bevel, frame_ bend_ m, leg_ aspect, leg_ inset, leg_ radius, leg_ rake, leg_ taper, outline_ exponent, pedestal_ base_ aspect, pedestal_ base_ radius, pedestal_ radius, spider_ sweep, support, taper, top_ surface, top_ thickness, trestle_ spread |
| Cabinet | columns, fill, front, handle, levels, metal, open_ back, open_ fraction, panel_ m |
| Bookcase | columns, fill, front, handle, levels, metal, open_ back, open_ fraction, panel_ m |
| Plant | branching, crown_ skew, density, droop, leaf_ aspect, maturity, phyllotaxis, pot_ fraction, pot_ radius, pot_ taper |
| TrashCan | finish, lid, rim_ width, squareness, taper |
| Whiteboard | border_ m, marker_ count, mounts, tray, tray_ fraction, tray_ side |
| Display | aspect, bezel_ m, curvature_ radius_ m, lift_ fraction, panel_ m, stand, tilt_ radians |
| Laptop | aspect_ ratio, bezel_ m, chassis_ m, corner_ m, hinge_ radians, keyboard_ columns, keyboard_ fraction, lid_ m, port_ count, speaker_ grilles, trackpad_ fraction |
| Monitor | aspect, bezel_ m, curvature_ radius_ m, lift_ fraction, panel_ m, stand, tilt_ radians |
| Notebook | binding, cover_ fraction, size_ variation, stack_ yaw, strap, volumes |
| Mug | fill_ fraction, style, taper, wall_ m |
| Books | binding, cover_ fraction, size_ variation, stack_ yaw, strap, volumes |
| Rug | clipped_ corners, rug_ exponent, rug_ taper |
| WallArt | border_ width, mat_ width, picture_ inset, shadow_ gap, style |
| Clock | bezel_ fraction, dark_ dial, digital, second_ hand, seconds, square |
| FloorLamp | shade_ height, shade_ radius_ fraction, shade_ sides, shade_ taper, stem_ offset, support |
| Keyboard | deck_ fraction, key_ area_ fraction, keyboard_ columns, keyboard_ rows, keyboard_ split, row_ stagger, wrist_ rest |
| Mouse | button_ fraction, mouse_ crown, mouse_ roundness, mouse_ waist, trackball |
| WaterBottle | fill_ fraction, neck_ fraction, shoulder_ fraction, style |
| PenHolder | variant |
| Printer | construction, lid_ fraction, roundness, split |
| StorageBox | construction, handle_ span, lid_ fraction, roundness, taper |
| CoatRack | arms, branch_ height, construction, handle_ span |
| Bag | construction, handle_ span, lid_ fraction, roundness, taper |
| CoffeeCup | fill_ fraction, lid, style, taper, wall_ m |
| SodaCan | heel_ fraction, lid, neck_ fraction, style |
| Notepad | binding, leaf_ fraction |
| Pencil | barrel_ sides, binding, tip_ fraction |
| Microphone | microphone_ bend, microphone_ head, microphone_ lateral |
| Phone | bezel_ fraction, camera_ island, case_ fraction, lens_ count |
| WallOutlet | edge_ radius, finish, gangs, rocker_ angle, screws, style |
| LightSwitch | edge_ radius, finish, gangs, rocker_ angle, screws, style |

| Category | Recorded controls |
|---|---|
| SideTable | capsule_ blend, edge_ bevel, frame_ bend_ m, leg_ aspect, leg_ inset, leg_ radius, leg_ rake, leg_ taper, outline_ exponent, pedestal_ base_ aspect, pedestal_ base_ radius, pedestal_ radius, spider_ sweep, support, taper, top_ surface, top_ thickness, trestle_ spread |
| MobileCart | feet_ height, levels |
| DrawerUnit | drawer_ bias, drawers, feet_ height, handle_ fraction |
| Bench | backed, roundness, slats |
| Ottoman | feet_ height, roundness |
| RoomDivider | panel_ fraction, slats |
| DeskLamp | reach, style, tilt |
| CoffeeMaker | belly, neck, style, taper |
| Kettle | belly, style, taper |
| Speaker | body_ taper, control_ fraction, profile_ roundness, style, support_ fraction |
| Headphones | control_ fraction, profile_ roundness, support_ fraction |
| Stapler | body_ taper, control_ fraction, support_ fraction |
| TapeDispenser | body_ taper, control_ fraction, support_ fraction |
| DocumentTray | body_ taper, sections, support_ fraction |
| DeskOrganizer | body_ taper, sections, support_ fraction |
| TissueBox | box_ height, finish, fold, lean, opening, phase |
| Vase | belly, flutes, neck, style, taper |
| Bowl | belly, flutes, style, taper |
| DeskFan | style, support_ fraction |
| Projector | corner_ m, lens_ fraction, lens_ offset, vent_ count |
| ProjectionScreen | screen_ drop, screen_ frame |
| PowerStrip | corner_ m, socket_ count |
| Cable | variant |
| WallShelf | bracket_ style, shelf_ brackets, shelf_ span |
| NoticeBoard | note_ count |
| WallSconce | diffuser_ fraction, sconce_ height, sconce_ taper, sconce_ width |
| CeilingPendant | diffuser_ fraction, shade_ fraction |
| FloorOutlet | lid_ height_ fraction, lid_ width_ fraction |

# 8 Procedural material space

## 8.1 Production bindings and sampling protocol

The material atlas covers all 36 architectural/object surface roles, with the Floor role split into timber, carpet and stone, plus a dedicated rug program: 39 rows. Each row records 256 consecutive material seeds, for 9,984 recipe records. The finish selector is the production seeded selector, not a round-robin assignment. Six fixed illustration seeds (0, 1, 4, 19, 53, 101) supply 234 native PBR renders, matched normal renders and base-color, tangent-normal and roughness map previews. Seed four is included to illustrate frosted interior glass. The full consecutive cohort, rather than these illustrations, defines the reported frequencies; failed seeds are not replaced.

Each render uses the actual `IndoorMaterials` factory and `for_part` binding, including palette tint, texture structure group, UV transform, clearcoat, anisotropy and transmission. The studio reuses a native Bevy renderer. Texture maps are baked by the production CPU reference implementation; this study does not measure GPU-baking throughput. The rig uses fixed 1,400/250 lux directional lights with shadows, the same seed-zero procedural reflection environment, indoor capture exposure and a $40^\circ$ vertical field of view. A 0.46 m diameter sphere and a $0.46 \times 0.49$ m beveled panel expose curved highlights and flat detail. Leaves use production blades with normalized atlas UVs. The backdrop is identical across columns and reveals glass transmission. These are material probes, not representative furnished rooms.

The separate assignment audit constructs every primary-room object in 256 consecutive mixed-layout rooms at density 0.65, without humans: 19291 objects and 69874 material parts. Rug fields use four bounded dye/pile slots; their bindings retain upholstery. Every supported object finish must have a valid slot in $[0, 12)$ and resolve from its recorded finish seed; deliberately independent book-cover slots are retained. Per-object-category surface and slot counts are exported. This cohort checks application frequency; the studio does not estimate it. Architectural walls/floors, human skin/hair, and installation-specific facade privacy treatments are outside the object-assignment denominator. Human complexion/groom variation is not represented by these substrate probes.

## 8.2 Correlated controls, bounded palettes

Within a room, twelve finish slots share at most three texture structures per surface. Ceramic, Fabric, FabricAlt and Leather have three structures; Wood, WoodEdge, Leaf, LeafLight, LeafVariegated and Succulent have two. Other roles reuse one structure; rug fields have up to four dedicated dye/pile structures. Shared map handles bound synthesis/upload costs; independent room material seeds resample the underlying recipe. These are correlated palettes, not twelve independent textures. Painted Metal is dielectric; Chrome is conductive. Screen, phone, television, artwork, printed-paper and whiteboard maps come from their dedicated content programs. Container glass and liquids retain several fixed optical constants; their presence is not evidence of broad optical diversity.

The page pairs each image with its actual PBR scalars and seed, and exposes eight-bin observed parameter marginals and object-assignment histograms. Optional subprograms keep their own denominators. Recorded ranges include conditional controls and are not causal influence measurements. Raw base-color maps precede the material multiplier; roughness previews show the packed green channel before its scalar multiplier (or the scalar when no texture exists). Tangent normals are distinct from rendered view-space normals. The maps include production mip chains, while map previews show level zero. A fixed RGB image or parameter count cannot establish perceptual volume, photographic realism, or 10M-sample learning utility. No product-of-ranges diversity estimate is reported.

## 8.3 Role inventory and realized object use

Table 11: Current material roles. Structures are the maximum shared map programs per room. Object parts and occupied finish slots are observed in the 256-room assignment cohort; zero means absent from that object-only cohort, not an unsupported material.

| Surface / floor type | Structures | Object parts | Slots seen |
|---|---|---|---|
| Paint | 1 | 0 | 0 |
| Accent | 1 | 0 | 0 |
| Wood | 2 | 3437 | 12 |
| WoodEdge | 2 | 4930 | 12 |
| FloorTimber | 1 | 0 | 0 |
| FloorCarpet | 1 | 0 | 0 |
| FloorStone | 1 | 0 | 0 |

| Surface / floor type | Structures | Object parts | Slots seen |
|---|---|---|---|
| Ceiling | 1 | 0 | 0 |
| Metal | 1 | 10908 | 12 |
| Chrome | 1 | 3239 | 12 |
| Plastic | 1 | 10756 | 12 |
| Fabric | 3 | 926 | 12 |
| FabricAlt | 3 | 2365 | 12 |
| Glass | 1 | 0 | 0 |
| Ceramic | 3 | 1222 | 12 |
| Soil | 1 | 1127 | 0 |
| Leaf | 2 | 715 | 12 |
| LeafLight | 2 | 706 | 12 |
| Paper | 1 | 5988 | 12 |
| Screen | 1 | 727 | 12 |
| Ink | 1 | 7086 | 0 |
| Light | 1 | 339 | 0 |
| Concrete | 1 | 495 | 0 |
| Art | 1 | 2811 | 12 |
| Rubber | 1 | 6907 | 0 |
| LeafVariegated | 2 | 81 | 12 |
| Terracotta | 1 | 321 | 0 |
| Bark | 1 | 715 | 0 |
| GlassInterior | 1 | 0 | 0 |
| ContainerGlass | 1 | 382 | 0 |
| Liquid | 1 | 225 | 0 |
| Drink | 1 | 634 | 0 |
| PhoneScreen | 1 | 736 | 12 |
| PrintedPaper | 1 | 333 | 12 |
| Leather | 3 | 818 | 12 |
| Whiteboard | 1 | 212 | 12 |
| Television | 1 | 126 | 12 |
| Succulent | 2 | 331 | 12 |
| Rug | 4 | 276 | 4 |

### 8.4 Control families

**Paint, accent and ceiling** Correlated scene pigment; roller, stipple, knockdown and trowel application; deposit scale, coverage, pinholes, gloss and local roughness.

**Timber and bark** Growth-ring warping, longitudinal/transverse grain, cut width, knots, stain pigmentation, finish absorption, pores and coating. Floor planks additionally vary board/joint scale.

**Textiles and leather** Weave/knit structure, yarn count and crimp, dye/pattern, pile and lustre; leather cell scale/stretch, grain, wrinkles, wear and coating. Carpet/rug pile and dye fields use separate room-scale programs, while upholstery resolves individual yarn crossings. Floor installations vary bond, basket strips, cut tint and roughness.

**Concrete, stone, soil and terracotta** Aggregate/grain distribution, deposits, veins, inclusions, polish and relief. The FloorStone row samples mineral mixtures rather than treating every tile as marble.

**Ceramic** Body/glaze mixture, pigment, gloss, clearcoat, firing variation, speckles, pooling, pinholes and crackle. Tint and texture structure are separately bound by the production factory.

**Botanical surfaces** Blade-aligned venation, chlorophyll patches, variegation, wax and restrained diffuse transmission. Succulent cuticles use a separate tileable pore/bloom program without blade venation.

**Metal, plastics, glass and content** Conductive versus painted bases, roughness and directional reflection; glass index, tint/attenuation, thickness and clear/frosted optics; independently seeded procedural screen/print content. Some constants are intentionally fixed by role.




Figure 24: Native PBR material probes: Paint, Accent, Wood, WoodEdge, FloorTimber, FloorCarpet. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

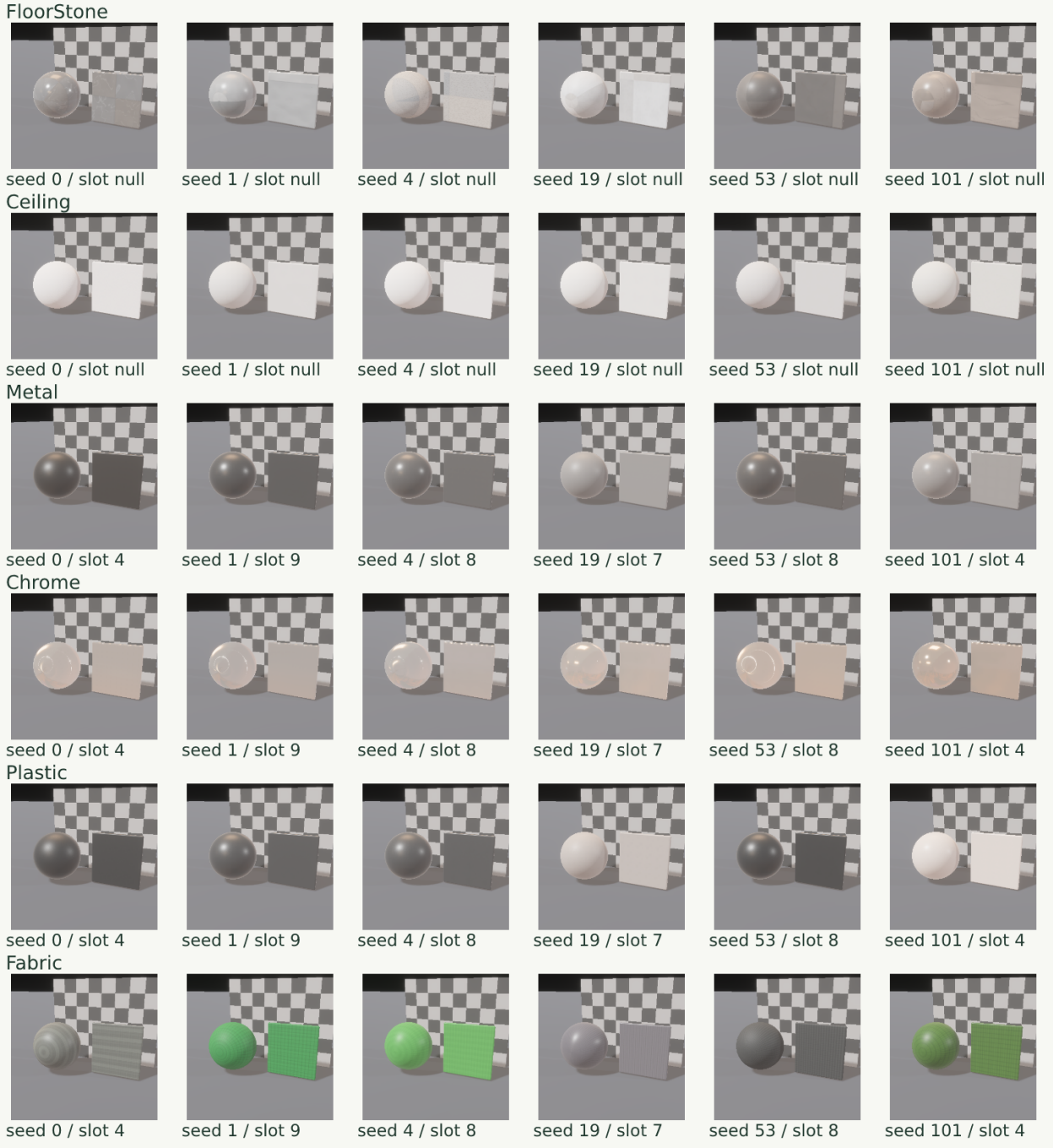



Figure 25: Native PBR material probes: FloorStone, Ceiling, Metal, Chrome, Plastic, Fabric. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

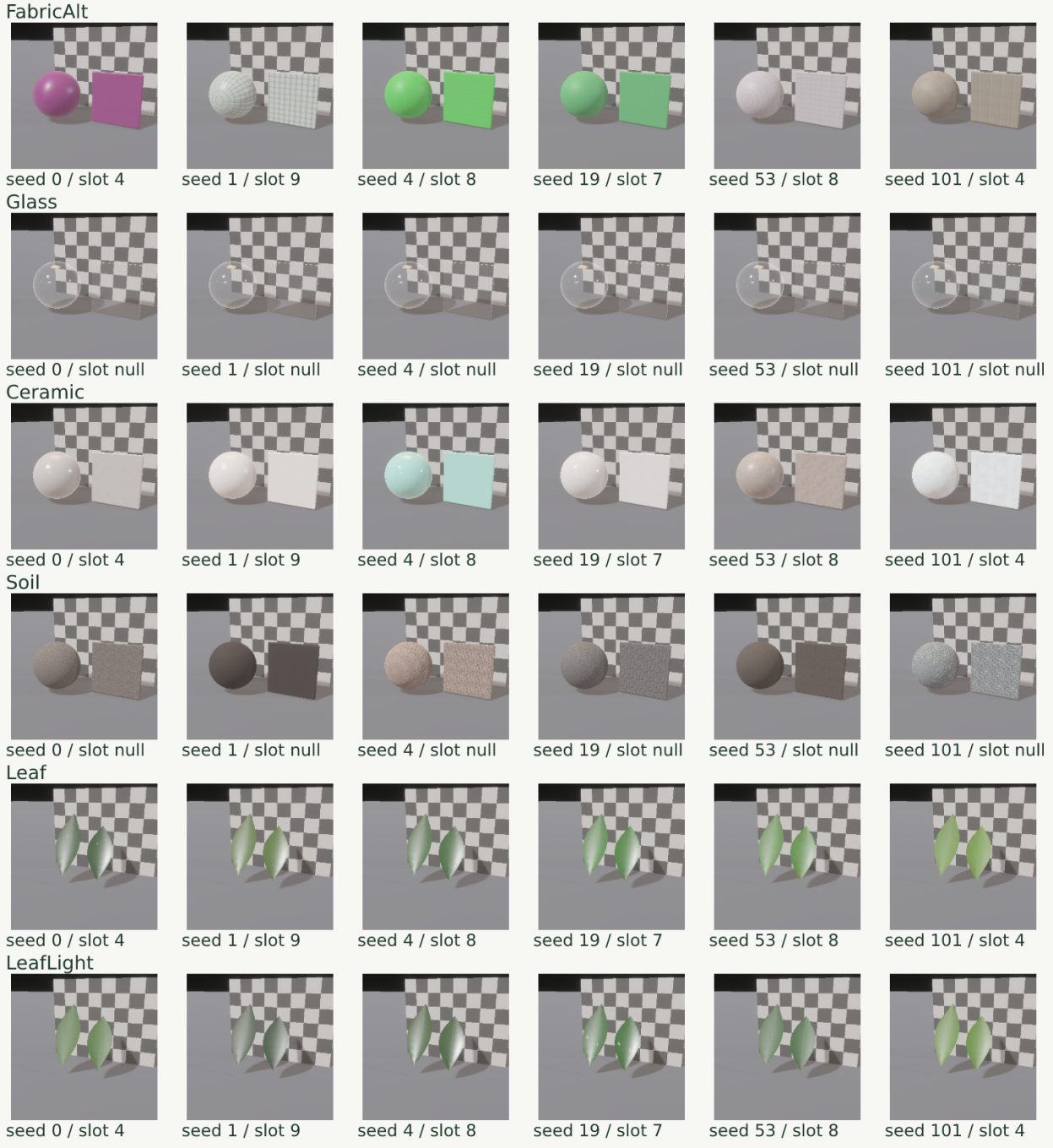



Figure 26: Native PBR material probes: FabricAlt, Glass, Ceramic, Soil, Leaf, LeafLight. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

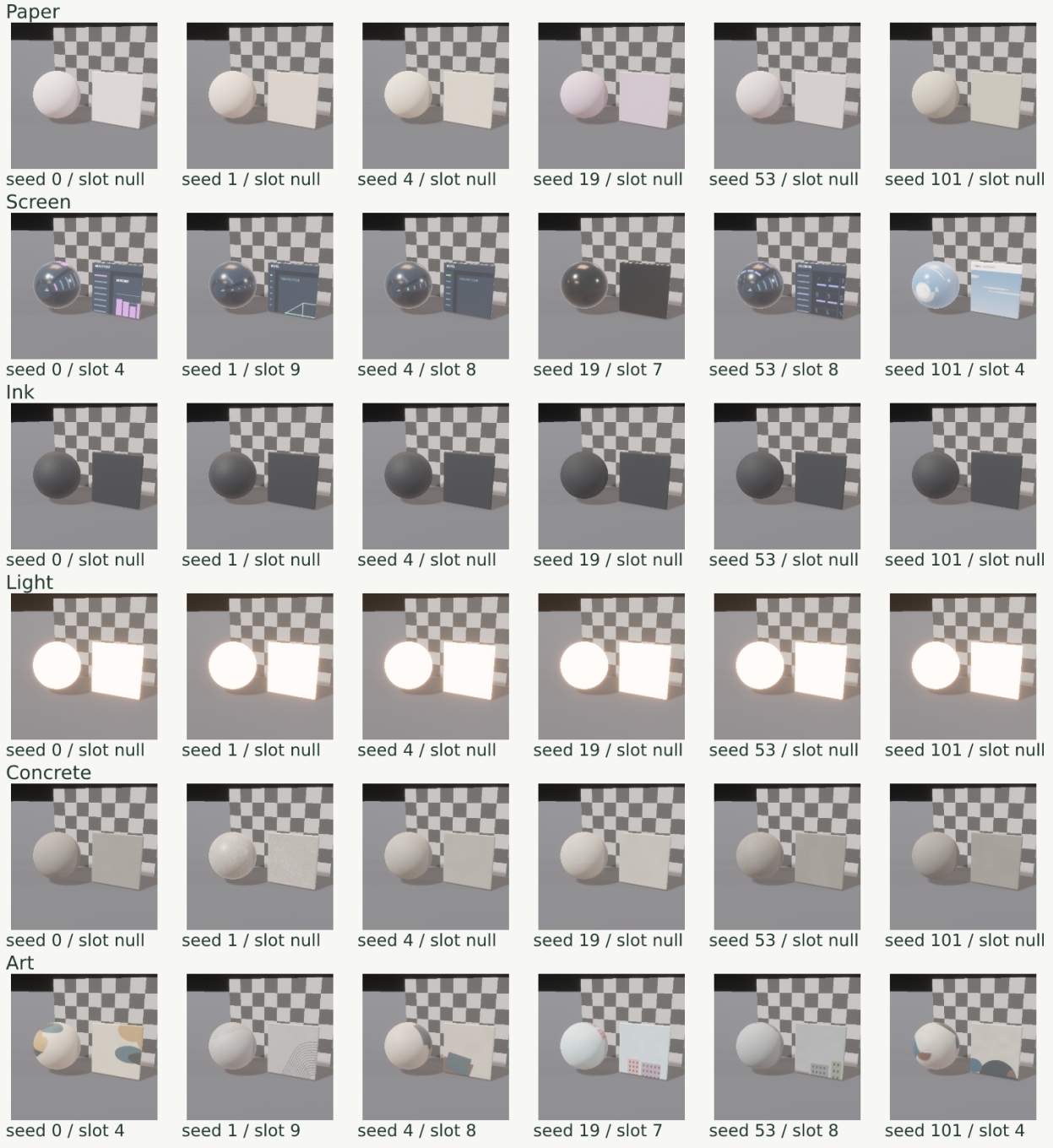



Figure 27: Native PBR material probes: Paper, Screen, Ink, Light, Concrete, Art. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

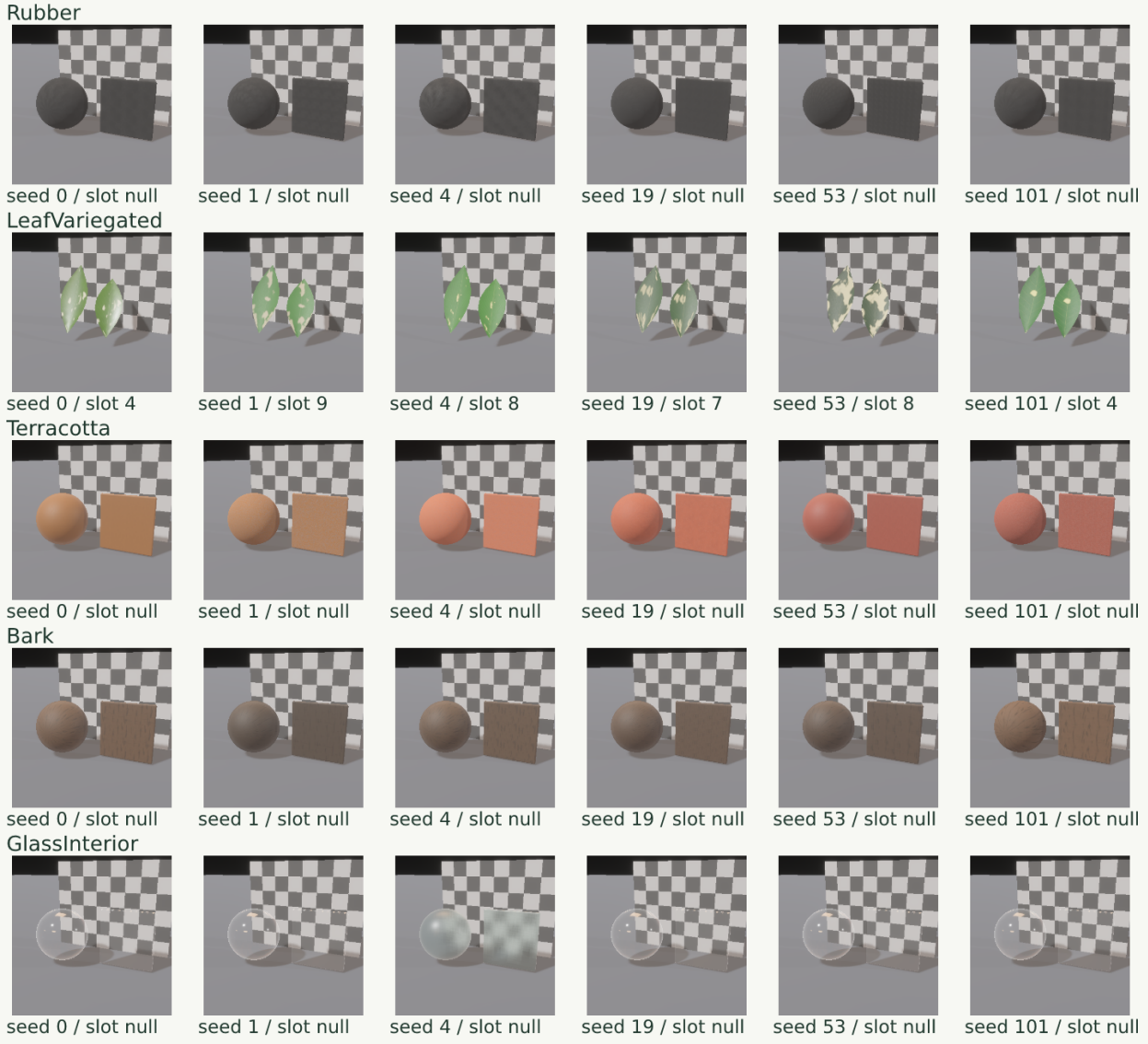



Figure 28: Native PBR material probes: Rubber, LeafVariegated, Terracotta, Bark, GlassInterior. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

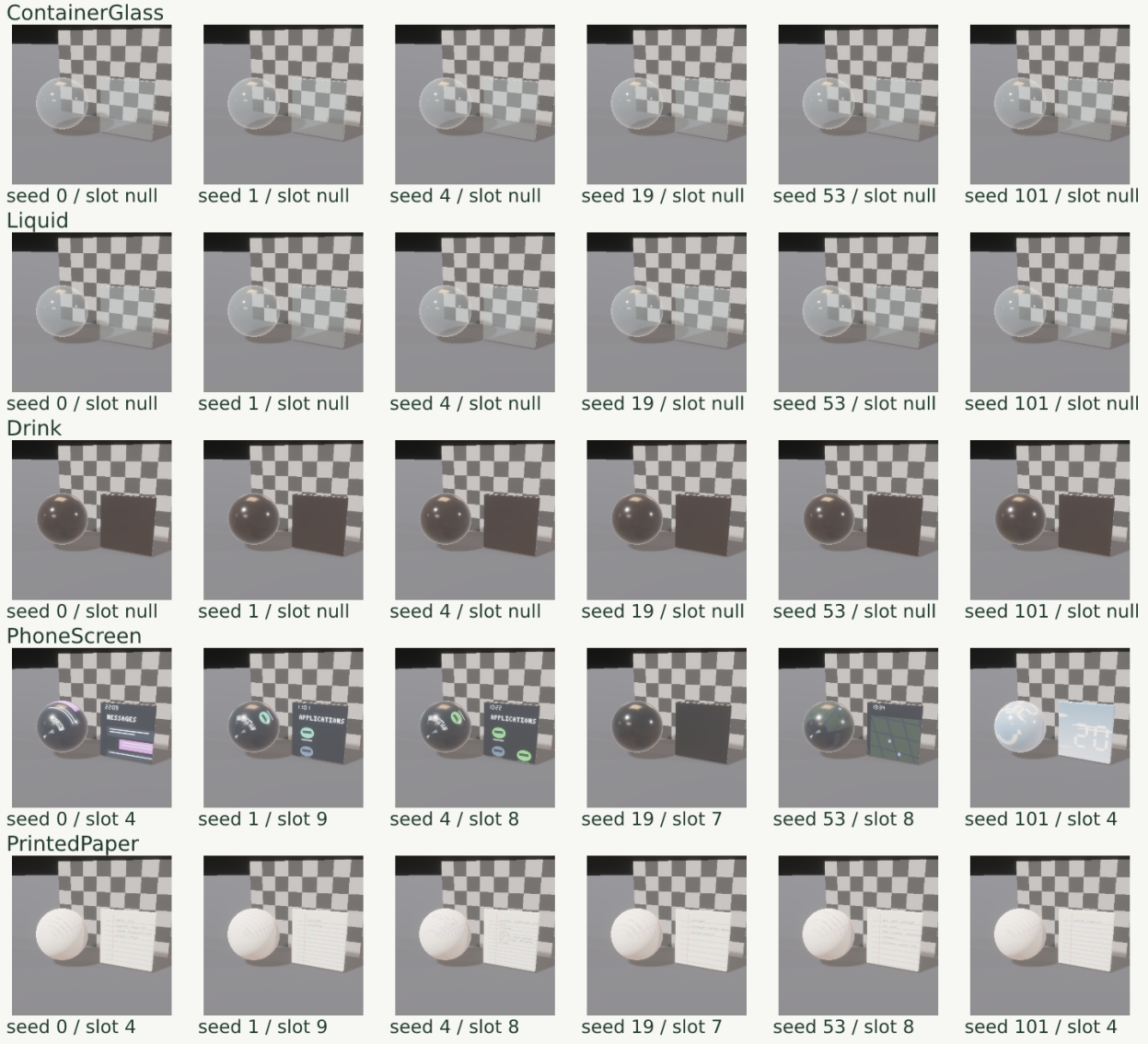



Figure 29: Native PBR material probes: ContainerGlass, Liquid, Drink, PhoneScreen, PrintedPaper. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

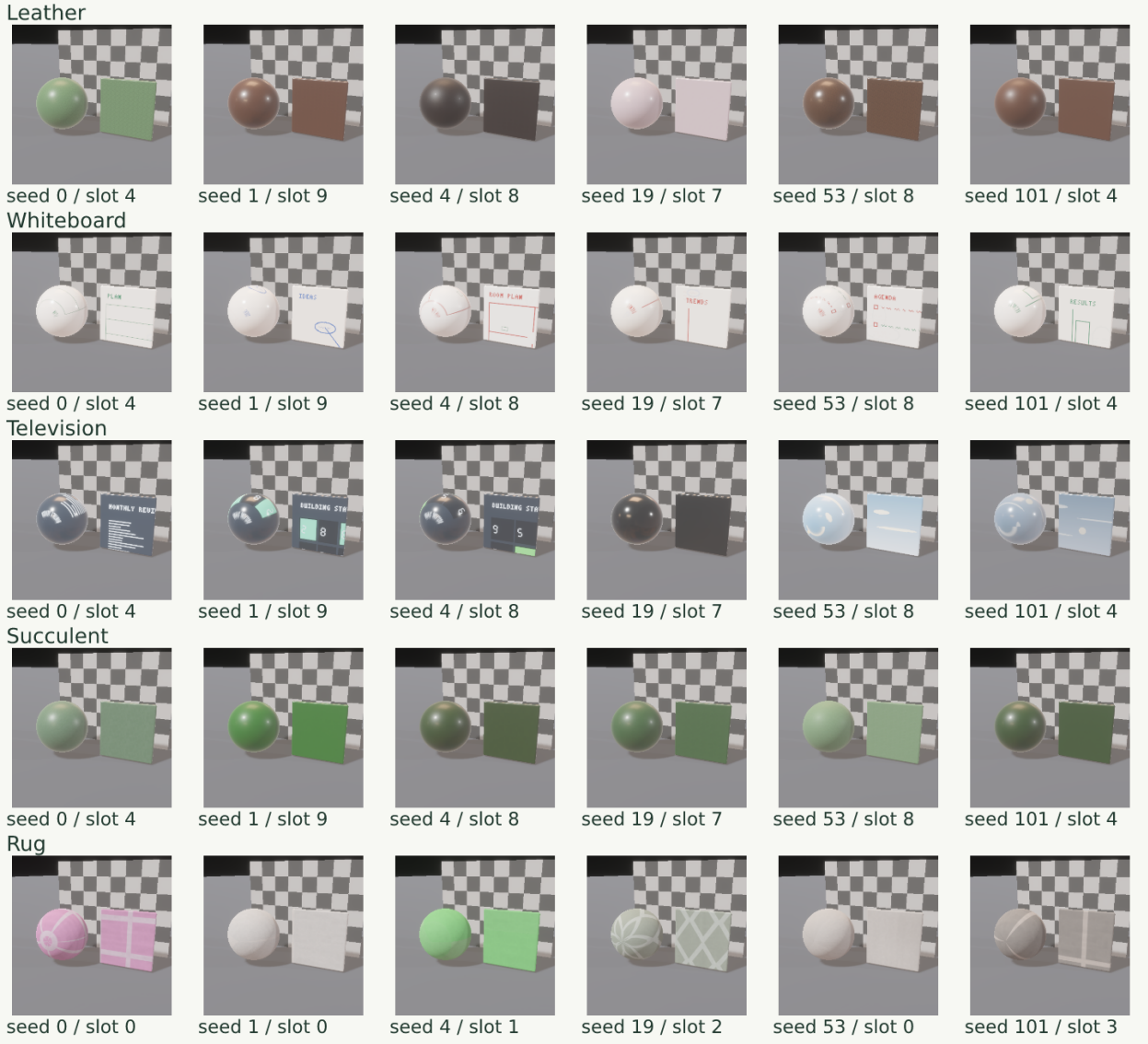



Figure 30: Native PBR material probes: Leather, Whiteboard, Television, Succulent, Rug. Columns share seed, camera, lights and geometry within each row; correlated production recipes and finish slots vary. Null slots use the base role. Six fixed examples illustrate, rather than estimate, the sampled distribution. Inspect the complete 256-seed marginals, map channels and actual PBR multipliers in the project-page atlas.

### 8.5 A parameter-linked example: ceramic

The following columns are the same six Ceramic images in the material plates. Recipe controls and final shader multipliers are shown separately. Glaze gloss drives the coating program; the roughness map retains local spatial detail. This is a correlated draw, not six isolated gloss interventions.

| Material seed | 0 | 1 | 4 | 19 | 53 | 101 |
|---|---|---|---|---|---|---|
| Texture structure | 1.000 | 0.000 | 2.000 | 1.000 | 2.000 | 1.000 |
| Glaze gloss | 0.792 | 0.624 | 0.948 | 0.840 | 0.412 | 0.184 |
| Crackle strength | 0.000 | 0.000 | 0.135 | 0.230 | 0.000 | 0.000 |
| Relief (mm) | 0.059 | 0.024 | 0.052 | 0.059 | 0.020 | 0.017 |
| Shader roughness multiplier | 0.943 | 0.867 | 0.980 | 0.927 | 0.907 | 0.933 |
| Shader clearcoat | 0.206 | 0.256 | 0.419 | 0.257 | 0.179 | 0.051 |

Table 12: Recorded recipe values and actual PBR bindings for the Ceramic grid. Full seed records and observed ranges for every material category accompany the project page.

## 9 Limitations

Procedural correctness, broad parameter variation and efficient capture do not establish photographic realism. Hair is a surface groom, skin lacks a full subsurface model, some furnishings remain sparse, object motion uses approximate collision proxies, and diffuse GI uses probes and simplified transport. Glass and lighting are approximations rather than full path tracing. Physical transport accuracy and visual plausibility require independent path-traced reference studies; the camera-baseline sweep does not measure either.

The current study is a generator audit and a small capture cohort, not a ten-million-scene run. It does not establish long-run resident-memory bounds, balanced coverage of all rare architectural combinations, collision-free arbitrary motion, suitability for every reconstruction loss, or transfer to real photographs. Shared-view defaults deliberately correlate cameras within a room; train/validation splits should therefore be made by room seed, not by image. A downstream assessment should freeze a training budget and real evaluation set, compare matched data quantities and camera policies, and measure reconstruction/calibration quality alongside redundancy. The manifests and metrics make that experiment reproducible without claiming its outcome in advance.

## References

[1] D. Xie et al. LRM-Zero: Training Large Reconstruction Models with Synthesized Data. NeurIPS, 2024. https://arxiv.org/abs/2406.09371.

[2] Y. Chen et al. MVSplat: Efficient 3D Gaussian Splatting from Sparse Multi-View Images. ECCV, 2024. https://arxiv.org/abs/2403.14627.

[3] M. Tschannen et al. SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. 2025. https://arxiv.org/abs/2502.14786.